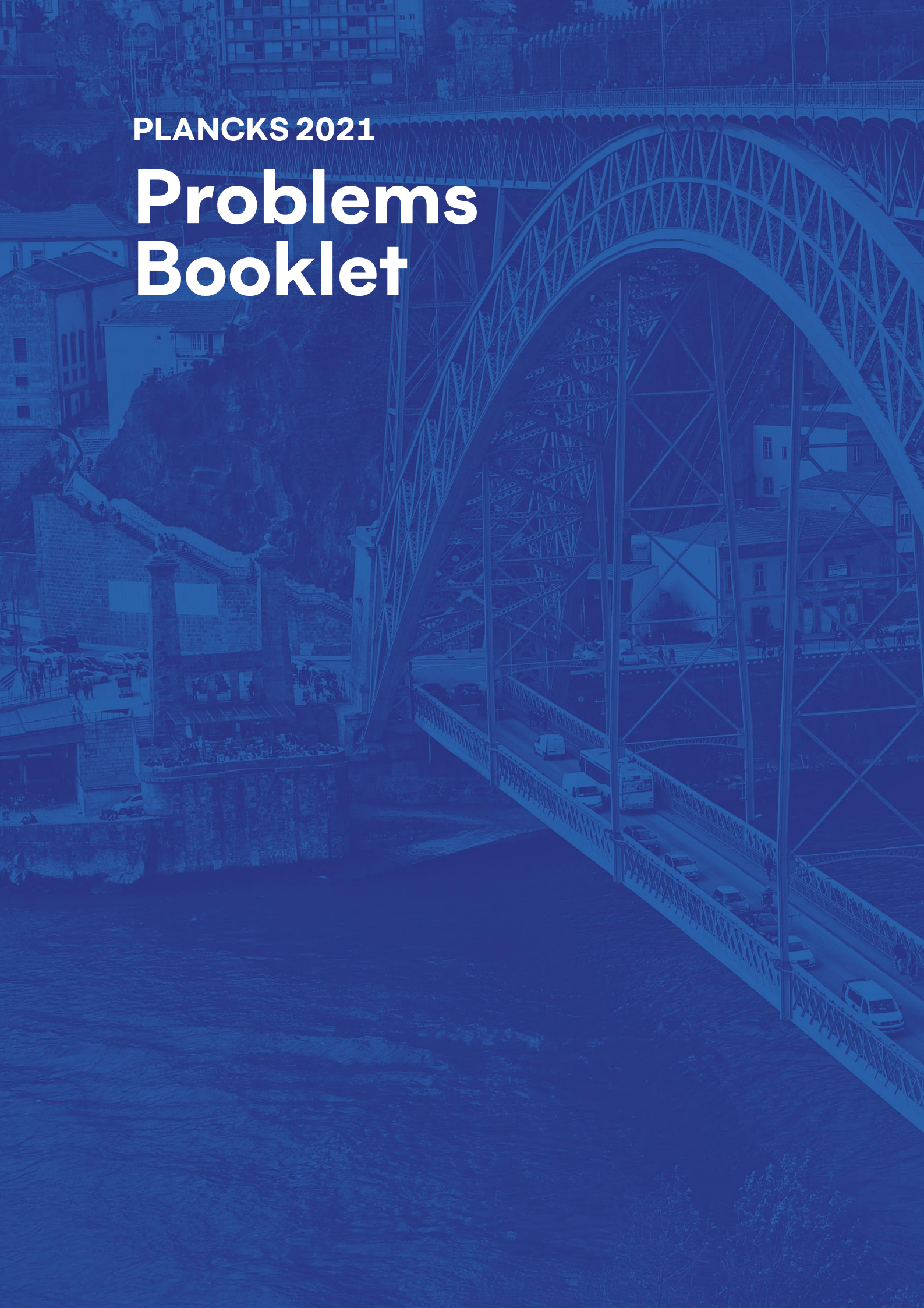
PLANCKS 2021

Problems Booklet



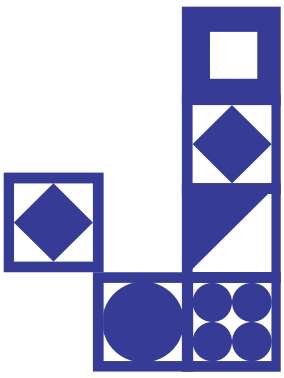
Rules

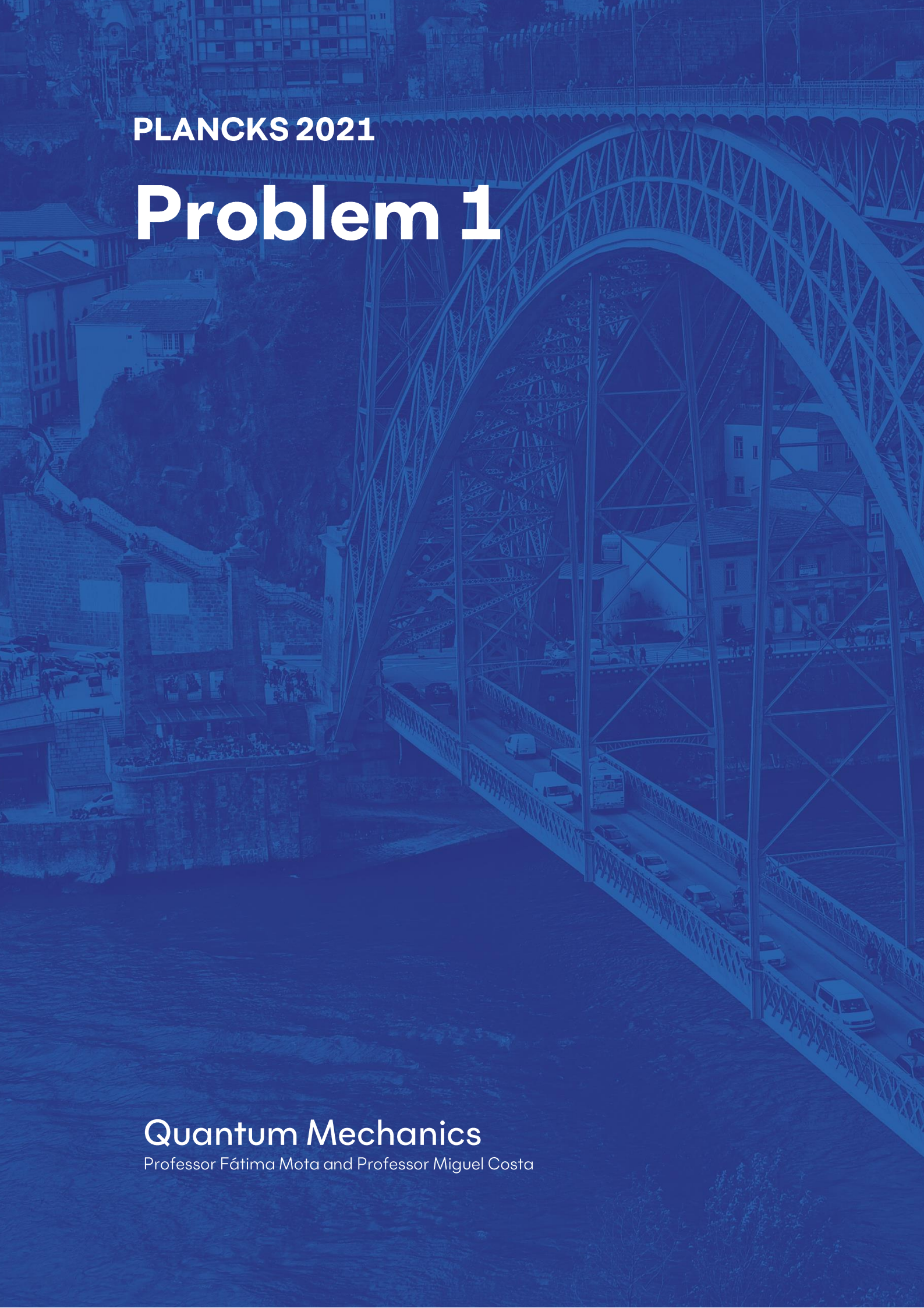
Dear contestants, Welcome to PLANCKS 2021!

Here are some rules and information you must read before starting to solve each problem:

1. The language used in the competition is English. If you use another language, it will not be considered.
2. The contest consists of 12 problems, each worth 100/1200 points. Subdivisions of points are indicated in the exercises.
3. All the exercises must be handed in separately and only one PDF per problem. For this purpose, you have different problem spaces in Moodle.
4. Please, when scanning your solution to a problem (you can use for instance CamScanner - Phone PDF Creator), write your team number on each page.
5. Make sure that your resolution is readable. Otherwise, the marking team has the right to not consider and reject the submission.
6. If you identify your team through other means that are not your team number, the submission will be rejected.
7. When a problem is unclear, a participant can ask, by tagging a @Supervisor - Problem X on Discord (where X is the problem number) for a clarification. If the response is relevant to all teams, the OC will provide this information to all.
8. You have 16 hours to submit this problem.
9. All the resolutions must be submitted before Saturday, May 8th, 12:00 (GMT+1).
10. Books and other sources can be consulted during the competition.
11. The organisation has the right to disqualify teams for misbehavior or breaking the rules. So please play fair, because we will know if you have not.
12. In situations to which no rule applies, the OC decides.

Best of luck!
May the best physics team win!





PLANCKS 2021

Problem 1

Quantum Mechanics

Professor Fátima Mota and Professor Miguel Costa

Consider a charged particle of mass m and charge q placed in a 3-dimensional isotropic harmonic potential of frequency ω_0.

Question 1 [30 points]

Assume the particle is placed in a time dependent and spatially uniform magnetic field $\boldsymbol{B} = \boldsymbol{B}_0 \sin(\omega t)$. Using first order perturbation theory indicate the allowed transitions from the ground state, calculate the correspondent amplitude for transition probability and show how you could calculate the transition probability (you don't have to carry out this final calculation).

Question 2 [70 points]

Now consider a system of twenty non-interacting identical particles of mass m and spin 1/2, placed in such an harmonic potential.

a. [25 points] Determine the ground state energy of the system.
b. [45 points] Assume that a constant magnetic field $\boldsymbol{B}$ is applied. Show that the diamagnetic susceptibility is constant.

Useful information

Remember:

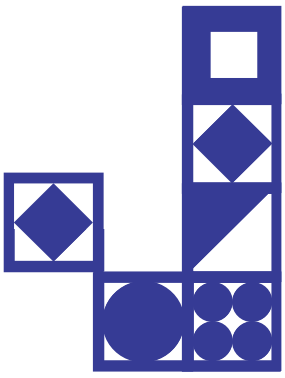
- minimal coupling procedure $\boldsymbol{p} \to \boldsymbol{p} - q\boldsymbol{A}$
- the vector potential for an uniform magnetic field may be written as: $\boldsymbol{A} = \frac{1}{2}\boldsymbol{B} \times \boldsymbol{R}$

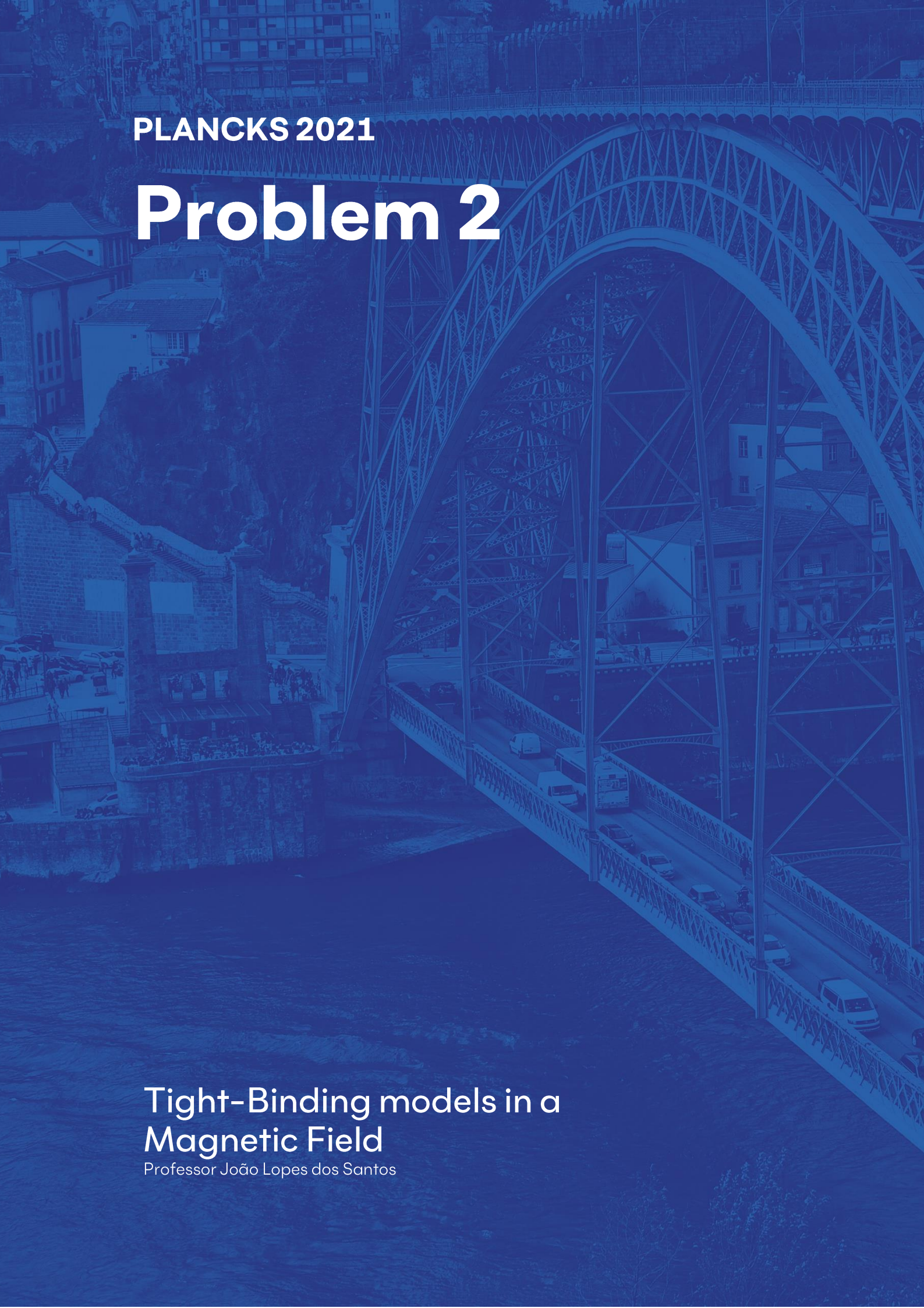
You may use the following information for the eigenfunctions of the 3-dimensional isotropic harmonic potential, in the $\{|n_x, n_y, n_z\rangle\}$ basis:

$$|n_x = 0, n_y = 0, n_z = 0\rangle = |k = 0, l = 0, m = 0\rangle$$

$$|n_x = 0, n_y = 0, n_z = 1\rangle = |k = 0, l = 1, m = 0\rangle$$
$$|n_x = 1, n_y = 0, n_z = 0\rangle = \frac{1}{\sqrt{2}}[|k = 0, l = 1, m = -1\rangle - |k = 0, l = 1, m = 1\rangle]$$
$$|n_x = 0, n_y = 1, n_z = 0\rangle = \frac{i}{\sqrt{2}}[|k = 0, l = 1, m = -1\rangle + |k = 0, l = 1, m = 1\rangle]$$

$$|n_x = 1, n_y = 1, n_z = 0\rangle = \frac{-i}{\sqrt{2}}[|k = 0, l = 2, m = 2\rangle + |k = 0, l = 2, m = -2\rangle]$$
$$|n_x = 1, n_y = 0, n_z = 1\rangle = \frac{-1}{\sqrt{2}}[|k = 0, l = 2, m = 1\rangle - |k = 0, l = 2, m = -1\rangle]$$
$$|n_x = 0, n_y = 1, n_z = 1\rangle = \frac{i}{\sqrt{2}}[|k = 0, l = 2, m = 1\rangle + |k = 0, l = 2, m = -1\rangle]$$
$$|n_x = 2, n_y = 0, n_z = 0\rangle = -\frac{1}{\sqrt{3}}|k = 2, l = 0, m = 0\rangle - \frac{1}{\sqrt{6}}|k = 0, l = 2, m = 0\rangle + \frac{1}{2}|k = 0, l = 2, m = 2\rangle + \frac{1}{2}|k = 0, l = 2, m = -2\rangle$$
$$|n_x = 0, n_y = 2, n_z = 0\rangle = -\frac{1}{\sqrt{3}}|k = 2, l = 0, m = 0\rangle - \frac{1}{\sqrt{6}}|k = 0, l = 2, m = 0\rangle - \frac{1}{2}|k = 0, l = 2, m = 2\rangle - \frac{1}{2}|k = 0, l = 2, m = -2\rangle$$
$$|n_x = 0, n_y = 0, n_z = 2\rangle = -\frac{2}{\sqrt{3}}|k = 2, l = 0, m = 0\rangle + \sqrt{\frac{2}{3}}|k = 2, l = 2, m = 0\rangle$$





PLANCKS 2021

Problem 2

Tight-Binding models in a Magnetic Field

Professor João Lopes dos Santos

Introduction

The problem of the spectrum of an electron gas in a magnetic field is addressed by the minimal coupling prescription

$$\frac{\nabla}{i} \to \frac{\nabla}{i} - \frac{q}{\hbar}\mathbf{A}(\mathbf{r}) \quad (1)$$

where $\mathbf{A}$ is the vector potential and the magnetic field is $\mathbf{B} = \nabla \times \mathbf{A}$. It is not difficult to show that in that an electron gas in two dimensions, for uniform $\mathbf{B}$ normal to the electron gas, one has a discrete spectrum of *Landau Levels*,

$$\epsilon_n = \left(n + \frac{1}{2}\right)\hbar\omega_c, \qquad \omega_c = \frac{eB}{m} \quad (2)$$

where ω_c is the classical angular frequency of electron orbits. Furthermore, each level has a degeneracy (not counting spin) equal to the number of flux quanta in the sample, i.e. equal to BA/ϕ_0 where A is the area and $\phi_0 = h/e$ the flux quantum.

In the presence of a lattice potential this problem becomes considerably more complicated.

A tight binding (TB) model assumes a local basis, with one or more states $|\phi_i\rangle$ in each lattice unit cell and is defined by two sets of parameters:

- local orbital energies: $|\phi_i\rangle\epsilon_i\langle\phi_i|$;
- hopping amplitudes between local orbitals: $|\phi_i\rangle t_{ij}\langle\phi_j|$;

The general state is defined by its amplitudes in this basis

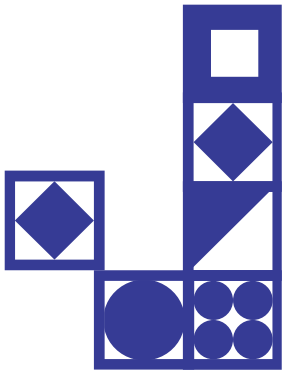
$$|\psi\rangle = \sum_i c_i\,|\phi_i\rangle \quad (3)$$

and for a crystalline lattice, Bloch's theorem can be used to solve for the Hamiltonian eigenstates.

The magnetic field is introduced by adding complex phases to the hopping amplitudes (Peierls substitution)

$$t_{ij} \to e^{i\phi_{ij}}t_{ij} \quad (4)$$

with the only requirement that the sum of phases along a loop is proportional to the magnetic flux in the area enclosed by the loop. The choice in Fig. 1 corresponds to a flux in the square of $Ba^2 = \phi$.



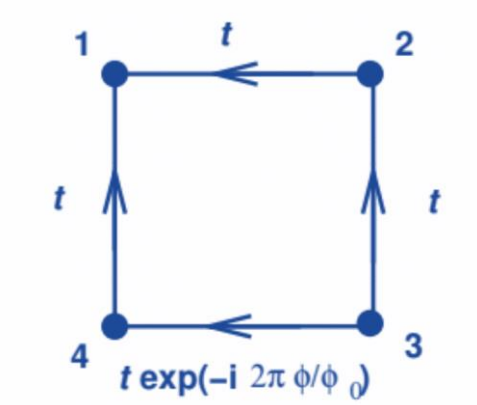


Figure 1 – With t real this choice of phase describes motion in a magnetic field perpendicular to the plane with flux in the square of $Ba^2 = \phi$

1. The square lattice

Assume a square lattice with sites $\mathbf{R}_{m.n} := a(m\hat{\mathbf{e}}_x + n\hat{\mathbf{e}}_y)$ with non-zero hopping amplitudes $-t$ $(t > 0)$ only between nearest neighbors. Take $0 \leq m \leq N_L - 1$ and $0 \leq n \leq N_W - 1$ ($N_W N_L$ unit cells). You may take the site energy to be zero (band center).

1.1. [20 points] You can make a choice of phases to represent a uniform magnetic field affecting only bonds along x, $\langle \psi_{m,n} | \mathcal{H} | \psi_{m\pm1,n} \rangle$. Go ahead and do so. Notice that you break translational invariance only along y direction. Use Bloch's theorem along x to reduce the Hamiltonian eigenvalue problem, for each Bloch wave vector k_x, to a 1D tight-binding chain with a on-site energy $V_n(k_x)$ that varies along the chain and is periodic. Determine $V_n(k_x)$.

1.2. [15 points] In realistic situation, the flux per unit cell, Ba^2, is much smaller that the flux quantum. Confirm this by estimating for $a \sim 1$Å the value of B such that $Ba^2 = \phi_0$. Show that the potential $V_n(k)$, for sensible values of B, has a wavelength much greater than the lattice spacing, a. Obtain a continuum limit for your TB equation by assuming that the tight binding amplitudes vary slowly with the index n and can be represented by a continuous function of y computed at the site coordinate $y_n = na$, which can be expanded in a power series. Show that for each k_x, you obtain a Schroedinger equation in 1D (along y) for a particle in a potential $V(y, k_x)$.

1.3. [15points] Look for solution close to the minimum of the potential. Reduce the problem to that of an harmonic oscillator and try to prove the following:

- The low energy spectrum takes the form

$$\epsilon_r = -4t + \left(r + \frac{1}{2}\right)\hbar\omega, \qquad r = 0,1,2 \ldots (5)$$

- if the potential period Ma is larger than the width $N_W a$ of the sample, each level has a degeneracy

$$p = N_W N_L / M;$$

- The form of the spectrum of Eq. 5 only holds for $r \ll M$.

2. The Graphene Lattice

The case of the graphene lattice brings further complications. The honeycomb lattice is not a Bravais lattice and has two carbon atoms per unit cell (see Fig. 2). The general state is

$$|\psi\rangle = \sum_{\mathbf{R}_{m,n}} a_{m,n}\,|\phi^{A}_{m,n}\rangle + b_{m,n}|\phi^{B}_{m,n}\rangle$$

where $\mathbf{R}_{m,n} = m\mathbf{a}_1 + n\mathbf{a}_2$ is a Bravais lattice site, and $|\phi^{A}_{m,n}\rangle$ and $|\phi^{B}_{m,n}\rangle$ are the two local orbitals in the unit cell at $\mathbf{R}_{m,n}$. In a minimal model the site energies of all local orbitals is taken to be zero and a non-zero hopping amplitude, $-t$, ($t \approx 3$ eV) exists only between nearest neighbors. The magnetic field can be introduced with phases affecting only the bonds connecting orbitals with the same n.

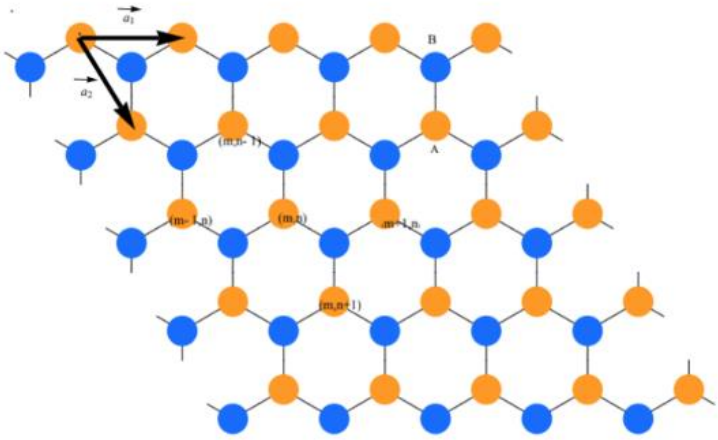


Figure 2- Graphene lattice with basic lattice translations

2.1. [20 points] Use Bloch's Theorem to reduce the eigenvalue problem to a 1D chain with two types of atoms with a space dependent hopping $t_n(k_1)$.

In graphene, the interesting energies are close zero. To obtain these states consider the following suggestions.

2.2. [20 points] Use a phase change of the local orbitals to reduce the TB equation of the AB chain to the form

$$\begin{aligned}\epsilon\tilde{a}_n(k_1) &= -t\left[2\cos\left(\frac{k_1 a}{2} - n\pi\frac{\varphi}{\phi_0}\right)\tilde{b}_n(k_1) - \tilde{b}_{n-1}(k_1)\right]\\ \epsilon\tilde{b}_n(k_1) &= -t\left[2\cos\left(\frac{k_1 a}{2} - n\pi\frac{\varphi}{\phi_0}\right)\tilde{a}_n(k_1) - \tilde{a}_{n+1}(k_1)\right]\end{aligned}$$

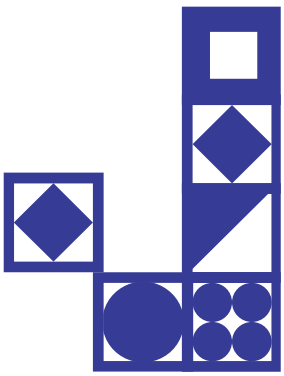
2.3. [10 points] You can obtain low energy states $|\epsilon| \ll t$, with slowly varying wavefunctions in an atomic scale, near values of n such that

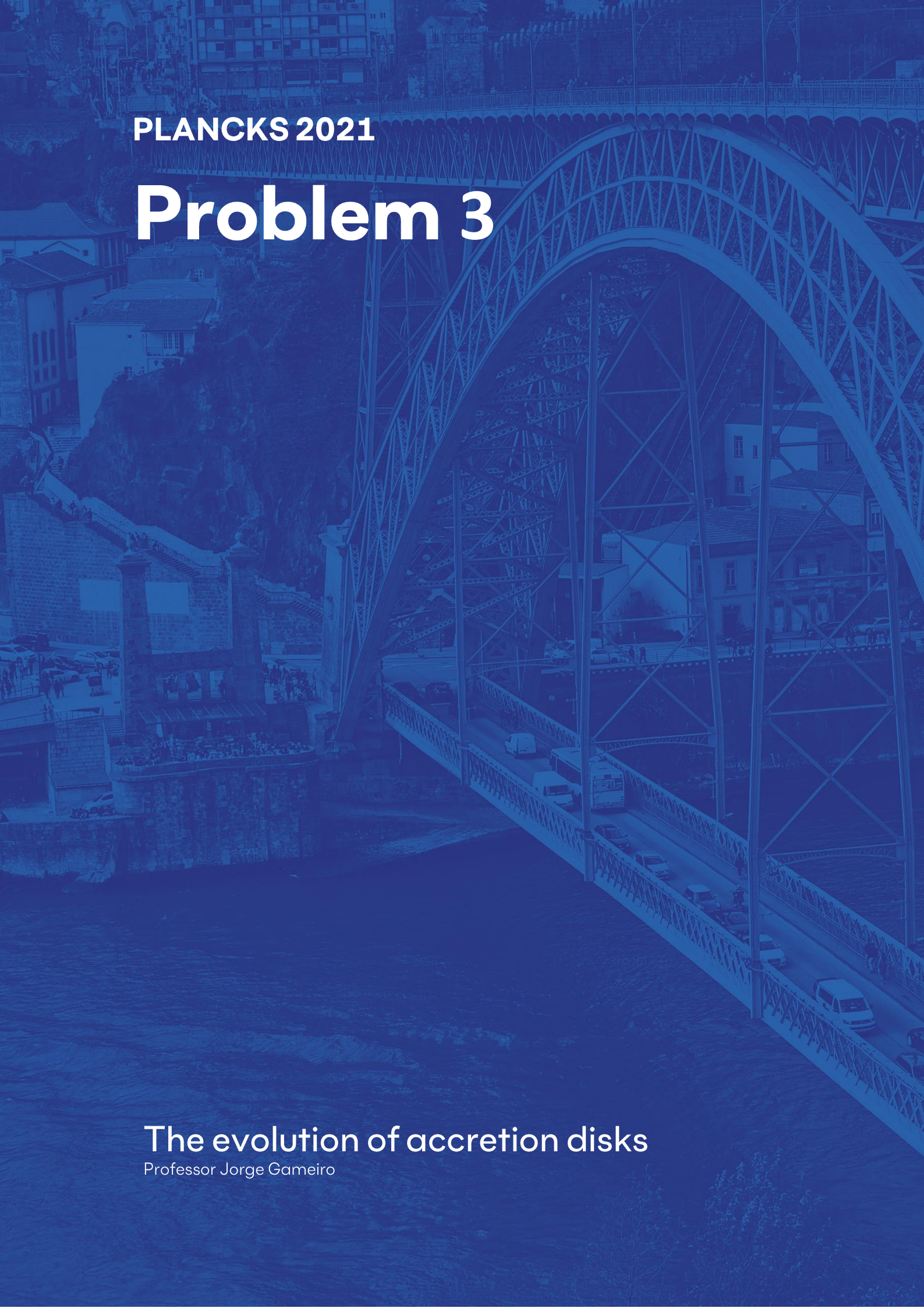
$$2\cos\left(\frac{k_1 a}{2} - \overline{n}\pi\frac{\varphi}{\phi_0}\right) = 1.$$

Expand the cosine term about $\overline{n}$, use a continuum aproximation for the amplitudes as you did for the square lattice, and obtain equations in the form

$$\begin{bmatrix} 0 & \hat{A}^{\dagger}(\partial_y, y) \\ \hat{A}(\partial_y, y) & 0 \end{bmatrix}\begin{bmatrix}\psi_a(y) \\ \psi_b(y)\end{bmatrix} = \epsilon\begin{bmatrix}\psi_a(y) \\ \psi_b(y)\end{bmatrix}$$

were the commutator $\left[\hat{A}(\partial_y, y), \hat{A}^{\dagger}(\partial_y, y)\right]$ is a c-number. Recall the commutation relation of harmonic oscillator operators, $[a, a^{\dagger}] = 1$, and try to figure out the low energy spectrum from this.





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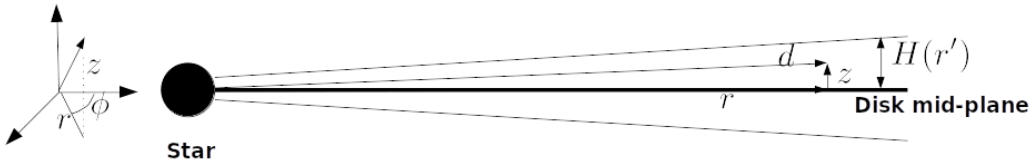
Problem 3

The evolution of accretion disks

Professor Jorge Gameiro

Introduction

The stars form through the gravitational collapse of dense molecular clouds. The specific angular momentum of gas in the molecular cloud typically matches the specific angular momentum of the gas in a circumstellar disk. In many cases the disk is confined so closely to the disk mid-plane that to a first approximation, one can assume the disk as a two-dimensional gas flow, the so-called thin disk approximation, where the thickness of the disk, $H(r) \ll r$.



Question 1 [20 points]

Assume for simplicity that the disk is optically thick and vertically isothermal, with a constant sound speed, c_s, and pressure given by $P(r,z) = \rho(r,z)c_s^2(r)$. If the gas is in hydrostatic equilibrium in the vertical z-direction (no mass motion in this direction), show that the vertical density profile, $\rho(z)$, is a gaussian profile. In this exercise one can consider the mass of the disk negligible when compared with the mass of the star, so the gravitational force is mainly due to the star.

Question 2 [30 points]

The evolution of a flat and geometrically thin disk follows from the equations of mass and angular momentum conservations. Consider a thin disk characterized by a surface density $\Sigma(r,t)$ (the mass per unit surface area of the disk, $\Sigma(r,t) \approx \rho(r,t) \times H(r)$), radial velocity $V_r(r,t)$ and angular velocity $\Omega(r)$.

The angular momentum conservation is given by the equation

$$r\frac{\partial}{\partial t}(r^2\Omega\Sigma) + \frac{\partial}{\partial r}(r^2\Omega.r\Sigma V_r) = \frac{1}{2\pi}\frac{\partial G}{\partial r} (1)$$

ν is the kinematic viscosity, G is the viscous torque exerted by the outer ring on the inner ring and has the form $G = 2\pi r.\nu\Sigma r\frac{d\Omega}{dr}.r$.

Get the mass conservation equation in cylindrical co-ordinates and together with equation 1 obtain the equation that represents the disk evolution (assume the angular velocity is keplerian),

$$\frac{\partial\Sigma}{\partial t} = \frac{3}{r}\frac{\partial}{\partial r}\left[r^{1/2}\frac{\partial}{\partial r}\left(\nu\Sigma r^{1/2}\right)\right] (2)$$

Question 3 [30 points]

In general, equation 2 is a nonlinear diffusion equation, because the kinematic viscosity ν may be a function of Σ, r and time. If we assume a constant viscosity ν in the full disk, the equation is linear.

Show that for a constant viscosity disk, the equation is a pure diffusion equation,

$$\frac{\partial f}{\partial t} = D\frac{\partial^2 f}{\partial R^2}$$

and determine the typical timescale of the disk evolution (also called the viscous timescale). [Hint: Consider $\mathrm{R} = 2\mathrm{r}^{1/2}$ and $\mathrm{f} = (3/2)\Sigma\mathrm{R}$]

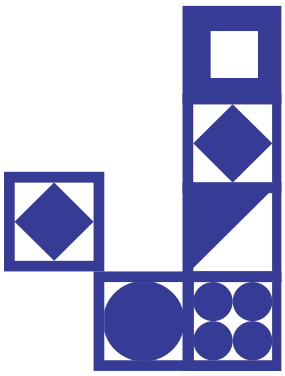
In your opinion, how could the disk evolution timescale be obtained from observations?

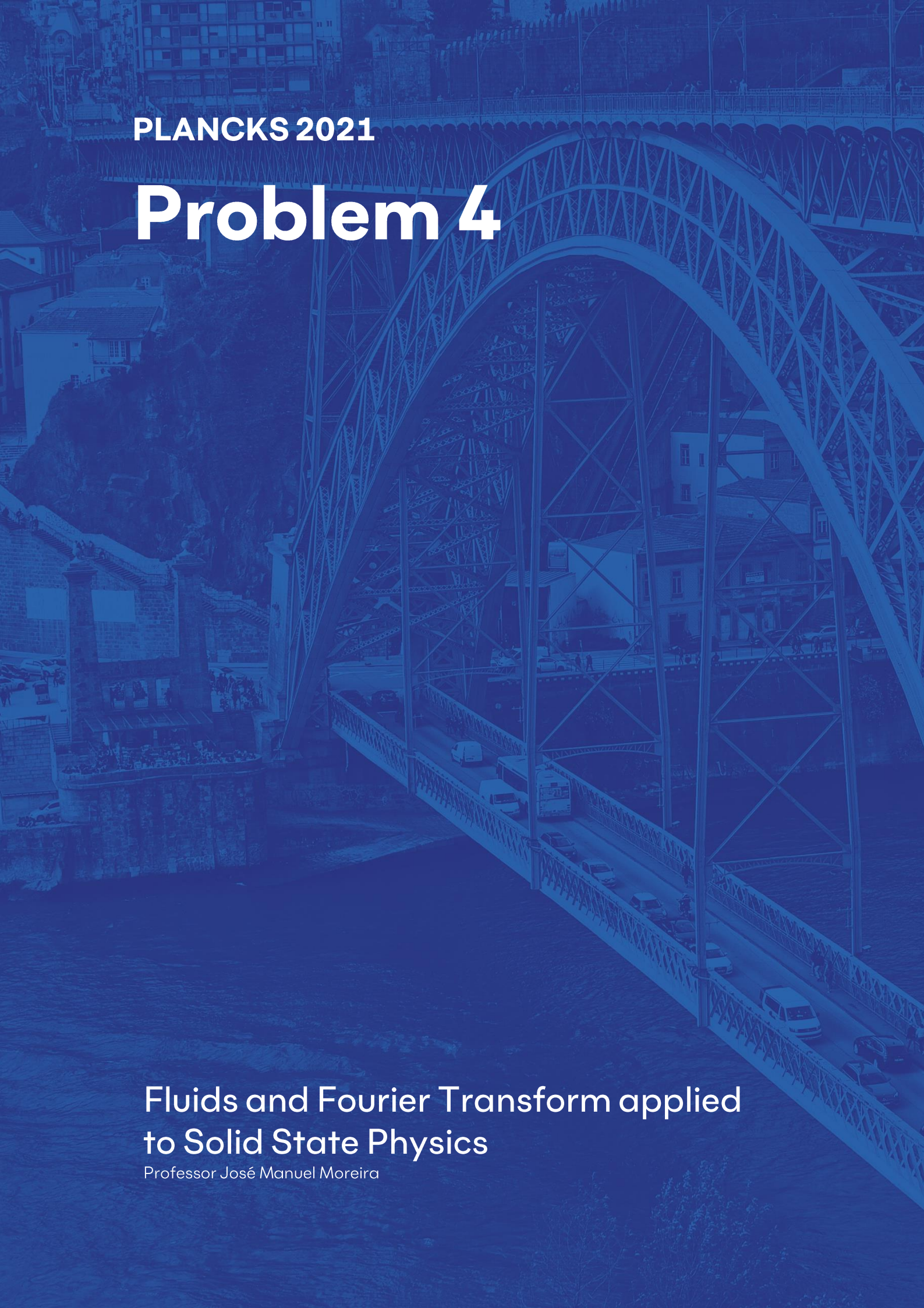
Question 4 [20 points]

In the previous questions we have assumed that the mass of the disk is negligible when compared with the mass of the star. Here, we want to discuss the validity of this assumption. As the disk is very large, for simplicity we approximate the disk by an infinite sheet with constant surface density Σ and thickness H. Show that the mass of the disk is not negligible when

$$\frac{M_{disk}}{M_*} > 0.5\left(\frac{H}{R_{disk}}\right)$$

[Hint: Determine the gravitational acceleration above the sheet due to the mass of the disk and compare this acceleration with the vertical component of the stellar gravity at $z = H$]





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Problem 4

Fluids and Fourier Transform applied to Solid State Physics

Professor José Manuel Moreira

1. An empty sphere inside a fluid

A spherical hole of radius R_0 (see Fig. 1) suddenly forms in a perfect incompressible fluid (specific mass ρ; weight negligible). The radius of the hole is small compared to the dimension of surrounding fluid.

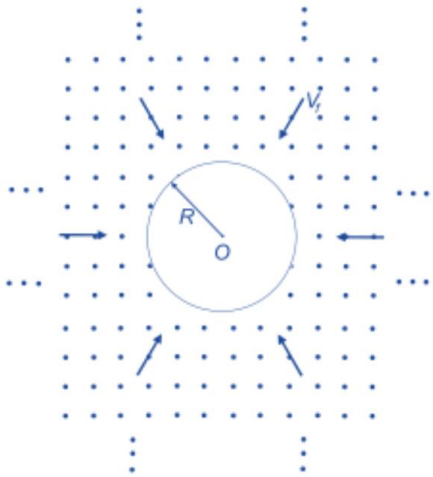


Figure 1- Spherical hole forming in a perfect incompressible fluid

1.1. [32.5 points] Prove that the time it takes for the fluid to completely fill the hole is

$$T = \sqrt{\frac{3\rho}{2p_0}} \int_0^{R_0} \frac{dr}{\sqrt{(R_0/r)^3 - 1}} \quad (1)$$

with p_0 corresponding to the pressure at "infinity".

1.2. [17.5 points] In order to calculate this time, we need to evaluate the integral:

$$\int_0^{R_0} \frac{dr}{\sqrt{(R_0/r)^3 - 1}} \quad (2)$$

Your task is to find the value of this integral in terms of beta or gamma functions. After that, choose your favorite numerical integration method and find the numerical value for this same integral and compare both. Hint: use the normalization $x = r/R_0$.

You can do the numerical iterations by hand or you can put your computer to do them for you, using a programming language like Python or a similar one.

2. Polarization field

Polarization is the vector field that expresses the density (C/m^2) of permanent or induced electric dipole moments in a dielectric material. When we apply an external electric field, the molecules of the material will acquire an additional electric dipole moment and the medium is said to be polarized. In a first approximation, the second order differential equation that relates the polarization of a material to the applied electric field can be written, in the time domain:

$$\frac{d^2P}{dt^2} + \gamma\frac{dP}{dt} + \omega_0^2 P - \nu\omega_p^2 P = \epsilon_0\omega_p^2 E(t) (3)$$

where $\omega_0^2 = \frac{e^2}{4\pi\epsilon_0 m R_0^3}$, $\gamma = \frac{1}{\tau}$, $\omega_\mathrm{p}^2 = \frac{\mathrm{Ne}^2}{\epsilon_0 \mathrm{m}}$ and N the number of charges per unit volume. In metals $\nu = 0$ and for an isotropic non-polar dielectric ν is theoretically $1/3$.

As we know, Fourier transforms, direct and inverse, and convolution integrals are one of the most valuable theoretical tools of physicists and engineers. They give us a clear correspondence between time and frequency domains. One can choose a domain to solve a problem (time or frequency domain) allowing us to do the math in the easier one. The equation 3 can be written in the frequency domain as:

$$\widetilde{P}(\omega) = \widetilde{R}(\omega)\,\widetilde{E}(\omega)$$

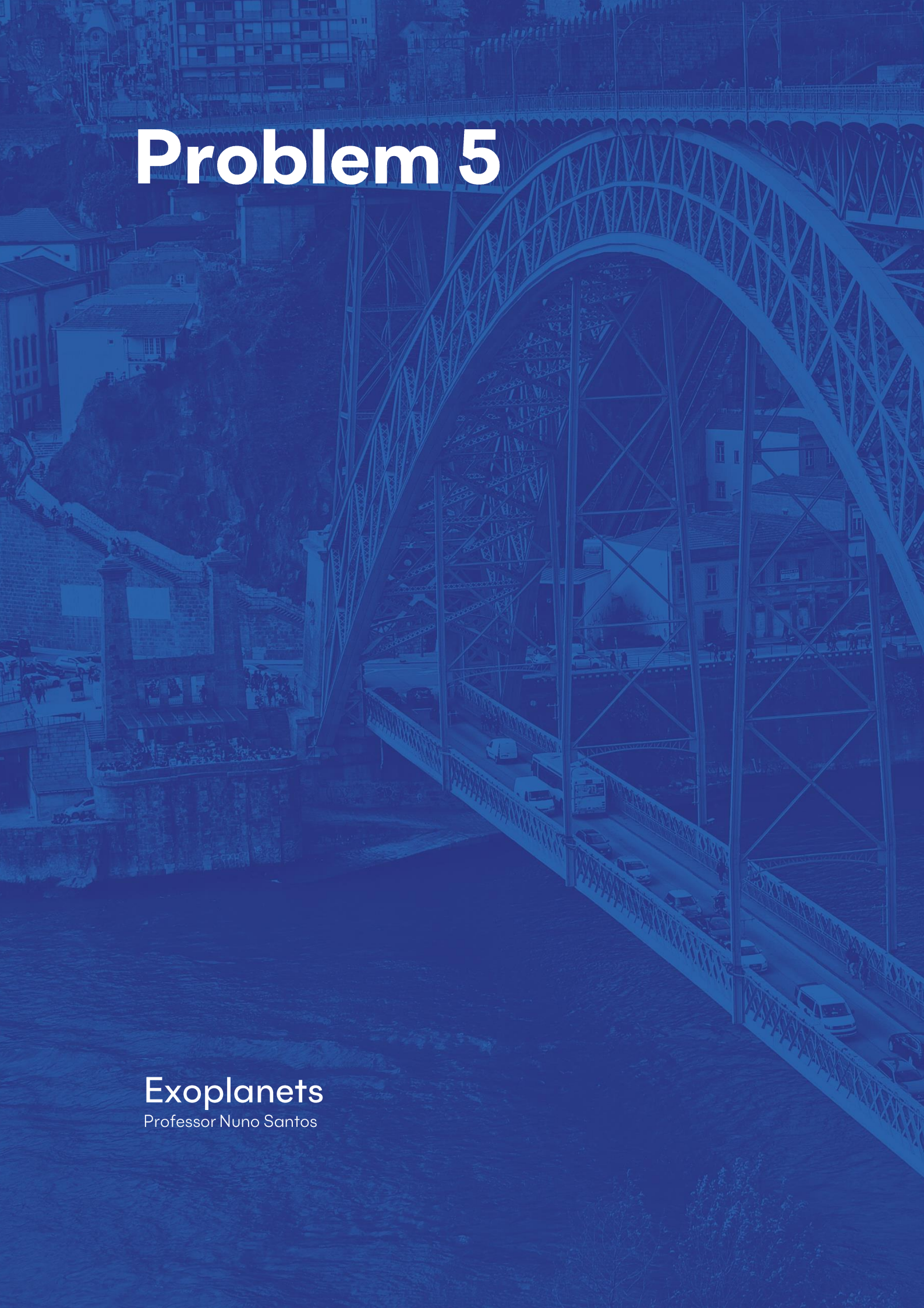
where $\widetilde{R}(\omega)$ is the transfer function (susceptibility) of the medium. In the general case, susceptibility is a tensor but in this exercise we consider it a scalar quantity.

2.1. [20 points] In the limit of $\gamma \to 0$, find expressions for the real and imaginary parts of $\widetilde{R}(\omega) = \widetilde{\chi}(\omega) = \chi_r(\omega) + i\chi_i(\omega)$.

2.2. [7.5 points] In this limit, find the response of the system to an applied electric field that is zero for $t < 0$ and E_0 for $t > 0$ (Heaviside function). Plot $P(t)$ and comment physically the result.

2.3. [22.5 points] If $\gamma \neq 0$, with $\frac{\gamma}{\omega_0} \ll 1$, comment on the amplitude of the resultant oscillation, its energy and corresponding oscillation frequency. Sketch the graphs for different values of increasing γ (always obeying the condition $\gamma \ll \omega_0$). What is the value of P when $t \to \infty$? Neglect irradiated electromagnetic energy and all other decay modes due to interactions with other particles.

Problem 5



Exoplanets

Professor Nuno Santos

Introduction

The 2019 Nobel Prize discovery of the planet orbiting a solar-like star marked, in 1995, the onset of a whole new area of modern astrophysics. Today, more than 4000 exoplanets have been detected orbiting other "suns". The results show that planets are ubiquitous in our Galaxy, but also that rocky planets are likely the most common among these. Complementary studies are allowing to characterize the planets in great detail. The measurement of accurate masses and radii (and thus mean densities) is setting strong constraints about their internal structure and composition. The detection of exoplanet atmospheres provides important clues about their nature and formation processes. The prospects of detecting and characterizing another Earth are now one of the main drivers for the development of new instruments and space missions by the main international agencies (ESO, ESA, NASA).

The following questions focus on some aspects of planet detection and characterization, but also on some interesting problems and challenges raised by planet formation models. All questions are independent and can be solved in any order.

Question 1 [25 points]

One of the main exoplanet detection and characterization methods (the so called Radial-Velocity method) is based on the measurement of the Doppler velocity of the star as it wobbles around the center of mass of the star-planet system.

Assume that you have a planet similar to Jupiter, with $\sim 10^{-3}$ solar masses, orbiting a Sun-mass star in a circular orbit. Assume that the orbital period is 12 years, and the orbital radius of the planet is 5.2 AU (Astronomical Units; 1 AU = 150×10^6km). Using simple principles, derive an expression that relates the Mass ratio (M_{star}/M_{planet}) with the orbital velocity ratio (V_{star}/V_{planet}) of the two bodies around the common center-of-mass.

Compute the expected orbital velocity of the star (in units of km/s).

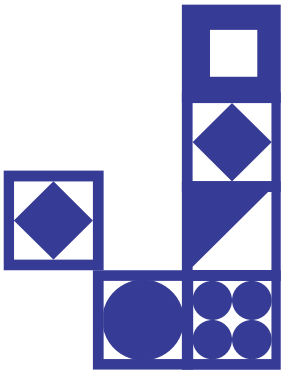
Question 2 [25 points]

Assuming that the planet is in thermal equilibrium with the star, and that the energy received by the planet is rapidly distributed on its surface (i.e., that the planet has uniform temperature), show that the temperature of the planet is proportional to $T_\star \sqrt{R_\star/D}$, where $T_\star$, $R_\star$, and D are the temperature of the star, its radius, and the distance between the star and the planet, respectively.

Comment on the physical nature of the proportionality factor.

Question 3 [25 points]

A significant fraction of the known exoplanets are giants, similar to Jupiter, but orbiting their host stars at very short distances. This raises several questions about planet formation and evolution, but also about the very existence of these worlds. Assume that the giant planet mentioned above is actually at 0.05 AU from a star similar to the Sun, such that its temperature is 1250 K.



Using simple principles, discuss if the atmosphere of this planet can survive against evaporation. For simplicity, assume that: 1) the atmosphere of a Jupiter-like planet is composed of hydrogen, 2) the planet has half the mass of Jupiter but the same average density (Data for Jupiter: M=1.898×10^{27} kg, R=70 000 km).

Question 4 [25 points]

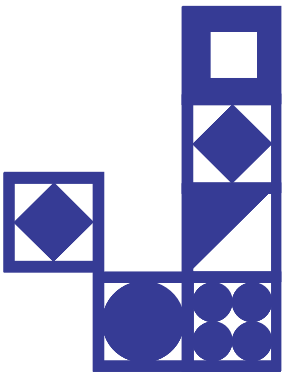
Planets are formed in disks of gas and dust that are formed as the outcome of the star formation process. In these disks, solids are expected to grow though collisions over timescales of several million years, eventually leading to planet size objects.

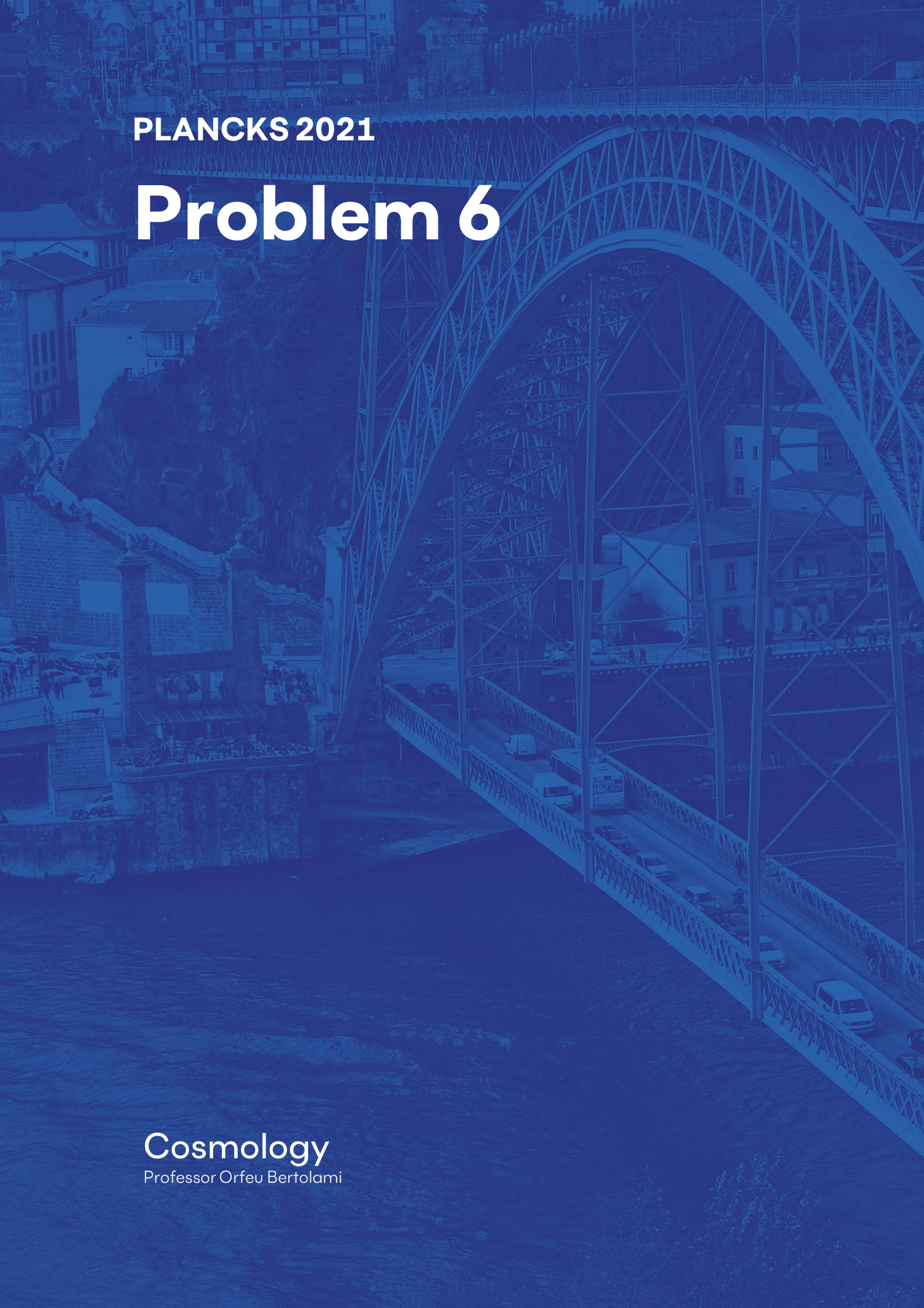
One of the biggest challenges of the planet formation process is related with the fact that, in a disk, the gas (that composes 99% of the mass of the disk) and the solids do not rotate at the same velocity. This leads to a gas drag that will make small, meter-sized pebbles to fall into the star in timescales of a few thousands of years.

Assume you have a disk of gas and dust whose density and temperature decrease with distance to the star. Show that, in such circumstances, you expect the velocity of the gas to be given by

$$v_{dust} \times \left(1 - \frac{c^2}{r^2\Omega^2}\right)^{1/2},$$

where $\Omega = \sqrt{GM/r^3}$ and c is the speed of sound.





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Problem 6

Cosmology

Professor Orfeu Bertolami

Introduction

The simplest cosmological model scientifically non-trivial can be obtained by Newtonian considerations assuming that a mass, M, is isotropically distributed in a volume, V_3, around an arbitrary origin. Suppose that, at a given time, a generic galaxy with mass m is at a distance $a(t)$ from the origin.

Question 1 [16 points]

Use Newton's Second law and the law of Universal Gravitation to obtain an equation of motion for $a(t)$.

Question 2 [16 points]

Show that this equation of motion admits the integral:

$$\frac{1}{2}\dot{a}^2 = \frac{GM}{a} - \frac{k}{2},$$

where k is an integration constant.

Question 3 [16 points]

Assume that the mass M is made up of homogeneously distributed pressureless dust with density, $\rho(t)$, within the radius, $a(t)$, of a sphere. Insert this mass into the previous equation to obtain an expression for the square of the expansion rate, $H = \dot{a}/a$, in terms of the density. This equation is known as Friedmann's equation and was obtained by the Russian polymath Alexander Friedmann (1888-1925) in 1922 in the context of the Theory of General Relativity, which means that this equation is more general than the Newtonian considerations assumed for the above derivation (small velocities, $v/c \ll 1$, and weak gravitational fields, $V/c^2 \ll 1$, where V is the gravitational potential).

Question 4 [16 points]

Assume that the mass M is constant to obtain a relationship between $\rho(t)$ and $a(t)$.

Question 5 [36 points]

In the Theory of General Relativity the integration constant, k, is associated to the spatial curvature of the Universe. There are three possible geometries to consider:

a. Closed Universe for $k > 0$, which corresponds to a Universe with a spatial geometry of a three-dimensional sphere, S^3, with radius of curvature, $R(t)$, related with the above model by

$$\frac{k}{a(t)^2} = \frac{1}{R(t)^2},$$

associated to the volume $V_3(t) = 2\pi^2 R(t)^3$;

b. Flat Universe for $k = 0$, which corresponds to a Universe with spatial geometry of a three-dimensional Euclidean space, E^3, for which $R \to \infty$;

c. Open Universe for $k < 0$, which corresponds to a Universe with spatial geometry of a three-dimensional hyperboloid, H^3.

Show that the evolution of the Universe can be understood for these three different cases rewriting Friedmann's equation as

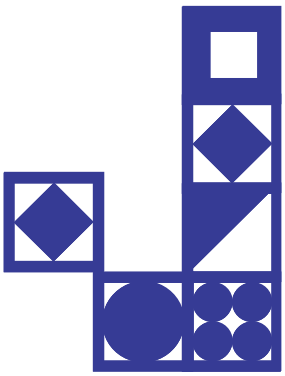
$$\frac{1}{2}\dot{a}^2 = -V_{eff}(a) - \frac{k}{2},$$

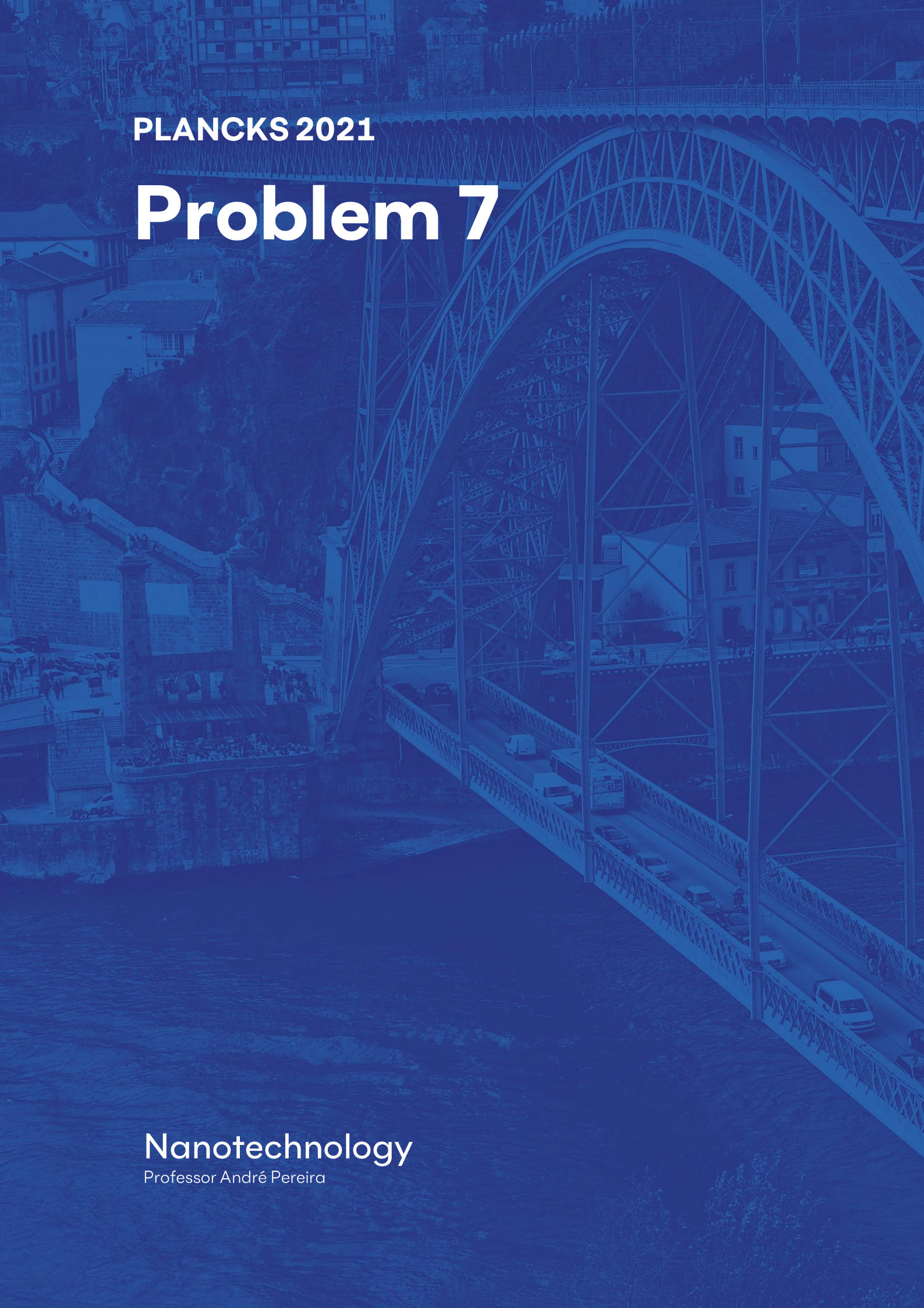
where $V_{eff}(a) = -4\pi G\rho a^2/3$, drawing the diagram of $V_{eff}(a)$ as a function of a and discussing how $a(t)$ evolves as a function of the cosmic time, t .

Relevant constants:

Newton's gravitational constant: $G = 6.67 \times 10^{-11}\ m^3\ kg^{-1}\ s^{-2}$

Speed of light: $c = 3 \times 10^8\ m/s$





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Problem 7

Nanotechnology

Professor André Pereira

Introduction

Nanotechnology is an emerging area that revolutionized the end and beginning of the XX and XXI Century, respectively. This field has already proved many advantages to improve the level of our society. In particular, Spintronics has a preponderant role in the growth and fast processors applied to several devices such as laptops, desktops, workstations, and servers.

One breakthrough was the invention of magnetic junctions constituted by the two electrodes that are ferromagnetic materials sandwiching a non-magnetic material, all at the nanometer thickness (Fig. 1). These devices can detect the binary unit (*bit*) in computing and digital communications, namely for information storage systems such as hard disk drive (HDD). The electric current in these devices consists of two partial currents in a ferromagnetic material, each with either spin-up or spin-down electrons. Moreover, they can present two different configurations related to the relative direction of the FM layers (these layers are usually metals).

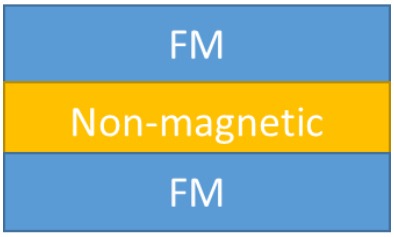


Figure 3- Two FM materials sandwiching a non-magnetic material

Question 1 [15 points]

Explain the working principle of these nanodevices and how they can measure a bit.

Question 2 [20 points]

Consider that the magnetic junction is a spin valve, e.g. the non-magnetic layer is a metal. What is the expression of the spin valve system's maximum sensitivity (magnetoresistance's maximum) using the simplest model? [Suggestion: consider the variation of resistance]

Question 3 [40 points]

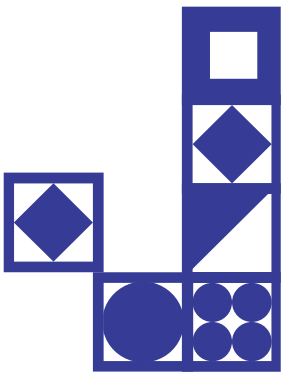
Consider now a magnetic tunnel junction barrier (MTJ) in which the non-magnetic layer is an insulator. Disregarding the magnetic contribution considering the metal layers, determine the tunnel current's general expression through the MTJ.

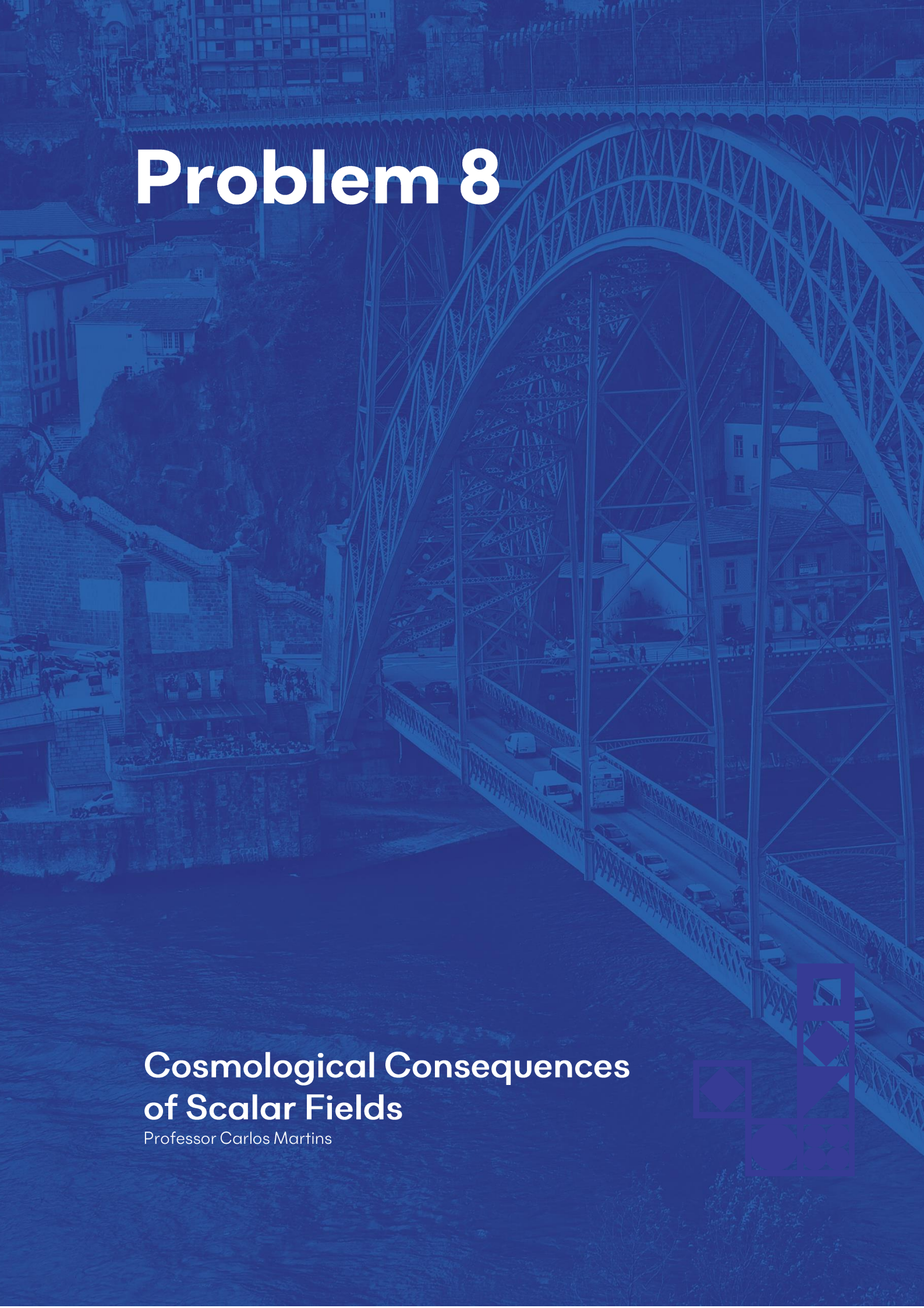
Question 4 [25 points]

Quantum sensors are another technology gaining extreme relevance. An example is the quantum dots nanoparticles that can be used as sensors for biomedical applications.

4.1 Determine the general expression between the bandgap (*Eg*) of quantum dots and its bulk counterpart.

4.2 What is expected for the quantum dot's *Eg* if the simple parabolic band's curvature presents a strongly curved parabola?





Problem 8

Cosmological Consequences of Scalar Fields

Professor Carlos Martins

Introduction

Since the 2012 discovery of a Higgs-like particle at the LHC, we know that fundamental scalar fields are among Nature's building blocks. Here we will explore some cosmological consequences of such scalar fields. We will assume homogeneous and isotropic universes, for which the Friedmann equation is

$$H^2 + \frac{k}{a^2} = \frac{8\pi G}{3}\rho\,, (1)$$

where a is the scale factor, $H = \dot{a}/a$ is the Hubble parameter (the dot denotes a time derivative), k is the curvature parameter, and ρ is the total density (a sum of those of the constituents of the universe). We will work in units where $c = 1$. It is also useful to know the continuity equation

$$\dot{\rho} = -3H(\rho + p) = -3H(1 + w)\rho\,, (2)$$

where p is the total pressure and for convenience we also introduced $w = p/\rho$, the equation of state parameter.

Question 1 [20 points]

Consider a scalar field with

$$\rho_1 = \frac{1}{2}\dot{\phi}^2 + V_1(\phi)\,, \quad p_1 = \frac{1}{2}\dot{\phi}^2 - V_1(\phi)\,, (3)$$

where V_1 is a generic potential. Calculate the cosmological evolution equation for this scalar field. Then repeat the calculation for a scalar field with

$$\rho_2 = \frac{V_2(\phi)}{\sqrt{1-\dot{\phi}^2}}\,, \quad p_2 = -V_2(\phi)\sqrt{1-\dot{\phi}^2}\,, (4)$$

where again V_2 is a generic potential.

Under what conditions can each field dominate the universe and cause its recent acceleration?

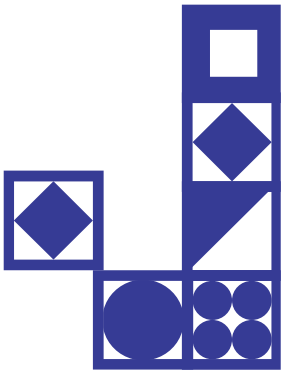
Question 2 [20 points]

Consider the first of the scalar fields in Question 1. Show that if the field speed is small one can write, to first order,

$$w(z) = -1 + (1 + w_0)\frac{H_0^2}{H^2(z)}\,, (5)$$

where w_0 and H_0 denote present-day values.

Further assuming a flat universe containing only matter and the scalar field (with present-day fractional contributions to the energy density Ω_m and Ω_ϕ), show that the Friedmann equation has the form



$$\frac{H^2(z)}{H_0^2} = \Omega_m(1+z)^3 + \Omega_\phi \left[\frac{(1+z)^3}{\Omega_m(1+z)^3 + \Omega_\phi}\right]^{\frac{1+w_0}{\Omega_\phi}} . (6)$$

Question 3 [30 points]

Consider a flat universe containing matter and a cosmological constant, but also a scalar field which obeys the cosmological evolution equation

$$\ddot{\phi} + 3H\dot{\phi} = -3H_0^2 \left[\eta_m \Omega_m \left(\frac{a_0}{a}\right)^3 + \eta_\Lambda \Omega_\Lambda\right], (7)$$

where η_m and η_Λ are constant coupling parameters describing how the scalar field couples to the matter and dark energy sectors and a_0 is the present-day value of the scale factor (you can assume $\phi_0 = 0$).

Assuming that you can neglect the scalar field's contribution to the Friedmann equation, solve the above evolution equation and obtain the explicit form of the redshift evolution of the scalar field.

Hint: Under the above assumption there is an exact analytic solution, which is easiest to obtain through a carefully chosen change of variables. You should obtain as final result a combination of logarithmic functions, which is typical for many cosmological scalar fields.

Question 4 [30 points]

An interesting observational consequence of scalar fields is that they lead to a variation of the fine-structure constant α (a measure of the strength of the electromagnetic interaction), and thus also to a violation of the Einstein Equivalence Principle (the cornerstone of General Relativity). For a universe containing only matter and a homogeneous scalar field (which is also responsible for accelerating the universe), α has the redshift dependence

$$\frac{\Delta\alpha}{\alpha}(z) \equiv \frac{\alpha(z) - \alpha_0}{\alpha_0} = \zeta \int_0^z \sqrt{3f_\phi(y)\left[1 + w_\phi(y)\right]} \frac{dy}{1+y} , (8)$$

where w_ϕ is the scalar field equation of state parameter,

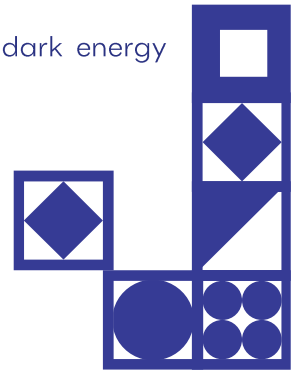
$$f_\phi = \frac{\rho_\phi}{\rho_m + \rho_\phi} (9)$$

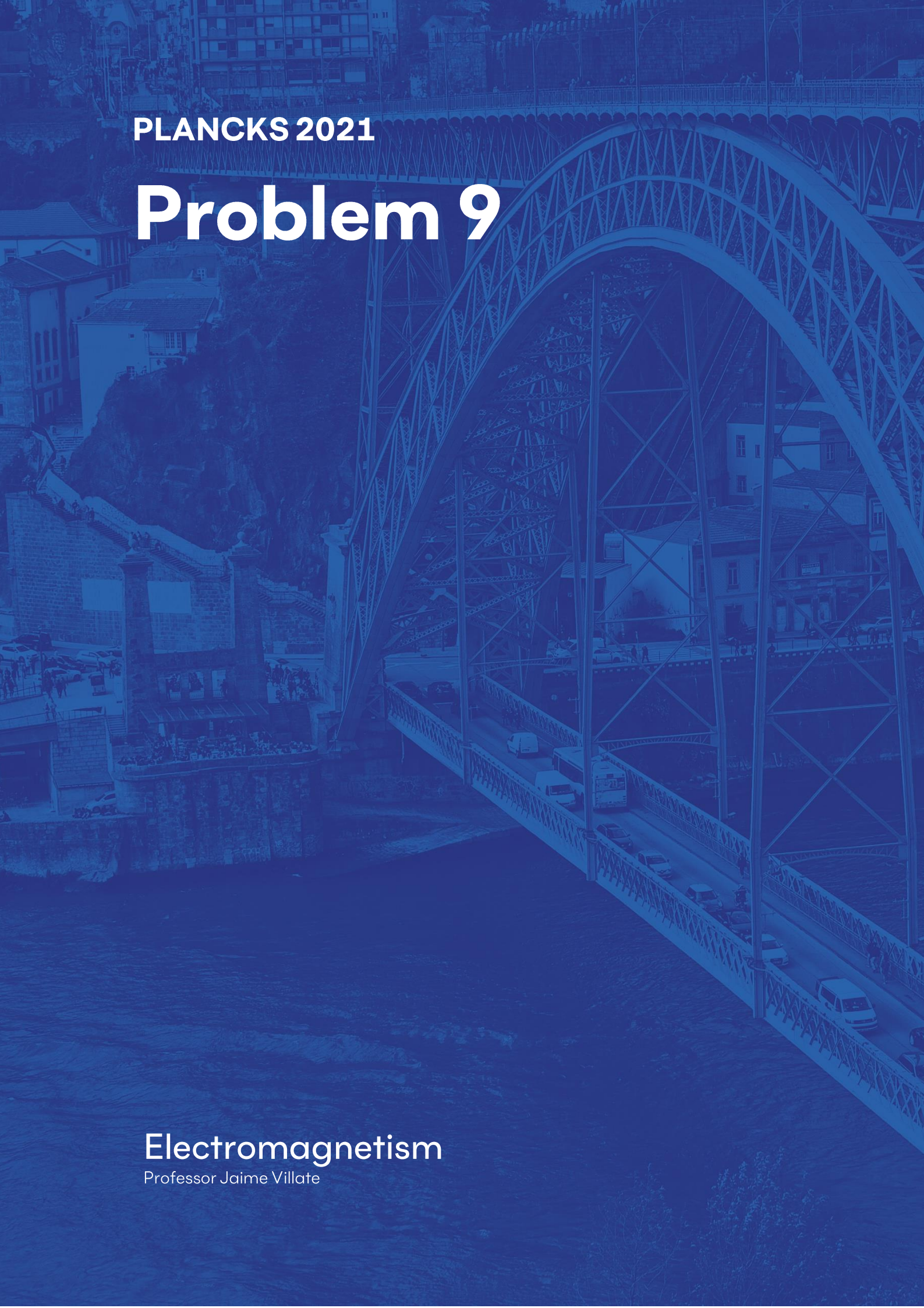
is the fractional contribution of the scalar field to the energy density of the universe, and ζ is another constant coupling parameter.

Calculate the generic explicit form (and redshift dependence) of the dark energy equation of state parameters that lead to a logarithmic dependence of α,

$$\frac{\Delta\alpha}{\alpha}(z) \propto \ln(1+z) . (10)$$

You may again assume a flat universe. Then calculate the explicit redshift-dependent form of the Friedmann equation for that generic equation of state parameter.





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Problem 9

Electromagnetism

Professor Jaime Villate

Introduction

A proton moves in a vacuum chamber, entering a region with a uniform magnetic field $\vec{B}$ that points into the page, as shown in the figure. The proton passes through point P, at a distance l from the region, with velocity $\vec{v}$, on the plane of the page and making an angle θ with the perpendicular to the border of the region. After entering the region, it exits at a point which is at a distance d from the point where it entered, and with velocity that makes an angle ϕ with the perpendicular to the border. Assume that v is large enough so the effect of gravity can be neglected during the trajectory, but much smaller than the speed of light making relativistic effects irrelevant.

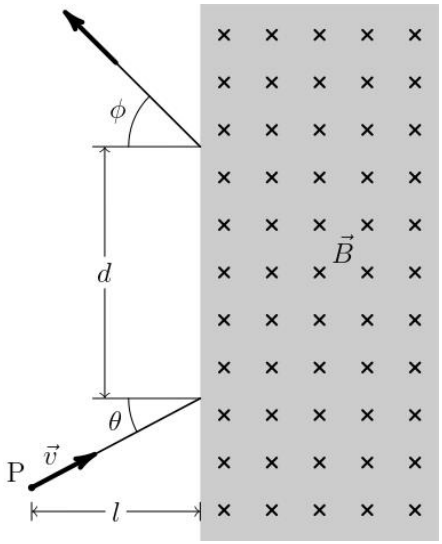


Figure 4 - Schematic representation of a proton entering a region with a magnetic field

Question 1 [10 points]

Find the exit angle ϕ.

Question 2 [20 points]

Find the distance d.

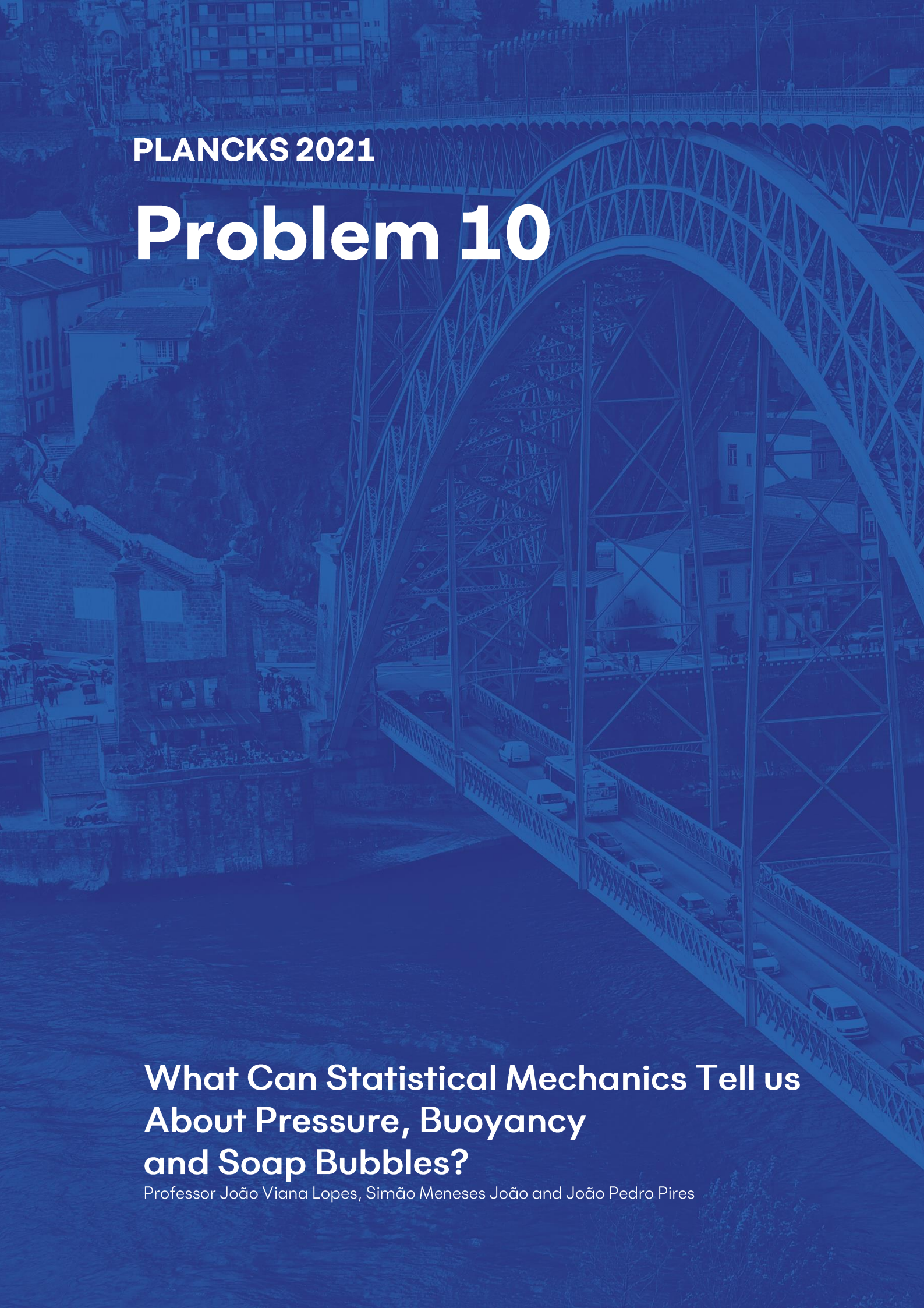
Question 3 [35 points]

If the direction of the uniform magnetic field is reversed, making it point out of the page, find the angle θ that would make the proton return to point P after exiting the region.

Question 4 [35 points]

With the initial magnetic field, pointing into the page, if the region is kept at a constant electrostatic potential V different from the

potential equal to zero outside the region, find the distance d.



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Problem 10

What Can Statistical Mechanics Tell us About Pressure, Buoyancy and Soap Bubbles?

Professor João Viana Lopes, Simão Meneses João and João Pedro Pires

Introduction

Albeit being a central quantity in the physics of fluids, hydrostatic pressure dresses itself in rather diverse definitions depending on the physical context. In thermodynamics, the pressure of a gas confined inside a container is defined in terms of the work done when the confining volume is changed. In contrast, in the kinetic theory of gases, pressure is seen as the rate at which free particles transfer momentum to the wall due to elastic collisions. In both cases, the very definition of pressure is a non-local one, that seemingly requires the introduction of a macroscopic hard-wall.

Meanwhile, in statistical physics thermodynamic state variables (e.g. internal energy, temperature, chemical potential, electric polarization, magnetization, etc...) can be described in terms of averages of microscopic observables with respect to an equilibrium probability distribution for the micro-state of the many-particle system. In order to define pressure in this language, one must find a way which does not depend on the existence of a container's wall, onto which the gas can exert force. Instead, we will consider a system of N independent particles in free space, in equilibrium at a temperature T and subjected to a general confinement potential – $V(\mathbf{r})$. Our aim is to provide a local definition of pressure in this setup using purely statistical arguments.

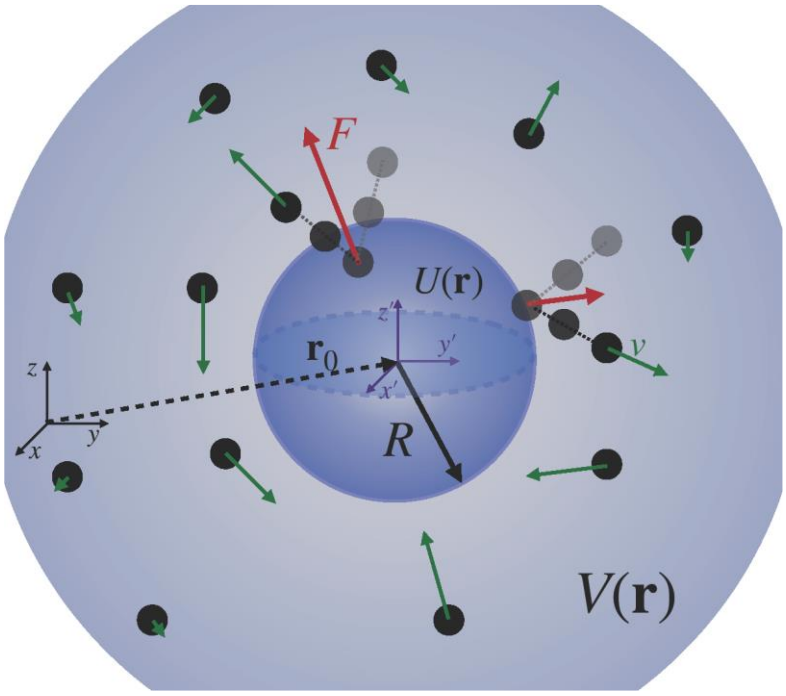


Figure 5 – Sketch of the physical situation.

1. Particle Density in Different Confining Geometries

The Hamiltonian for each particle of a non-interacting gas is simply

$$H = \frac{\mathbf{p}^2}{2m} + V(\mathbf{r})$$

for a general external confining potential $V(\mathbf{r})$.

1.1. [8 points] Using the canonical ensemble, show that the particle density is

$$\rho(\mathbf{r}) = Ce^{-\beta V(\mathbf{r})}$$

and determine the constant C.

1.2. Sketch and characterize the particle density profiles for the following confinement $V(\mathbf{r})$:

c. [4 points] A cubic box (of side L) with hard walls

d. [4 points] A semi-infinite cylinder with hard walls defined by $x^2 + y^2 \leq R^2$ and $z > 0$, with a gravitational potential mgz.

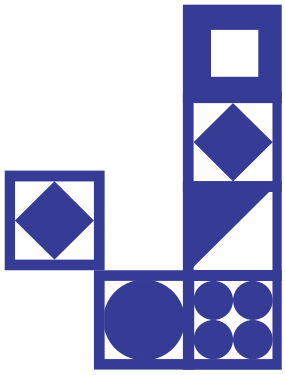
e. [4 points] A three-dimensional harmonic well $V(\mathbf{r}) = \frac{1}{2}k|\mathbf{r}|^2$.

2. Local Pressure as a Statistical Quantity

To define a local pressure, let us introduce a small perturbation to (probe) the gaseous system – a small (radius R) and immovable sphere centered in position $\mathbf{r} = \mathbf{r}_0$. With no loss of generality, we can use a coordinate system centered in the probing sphere $\mathbf{r}' = \mathbf{r} - \mathbf{r}_0$ to make the calculations easier. The independent particles will repeatedly bounce off this sphere, as shown in Fig. 1, transferring momentum in each collision. This is the physical origin of gas pressure. Actually, only the momentum component normal to the surface will contribute to this pressure. To calculate it from such a microscopic description, we take the sphere's surface, not as a rigid wall, but rather as a steep potential which allows the particles to penetrate, whilst repelling them strongly in the outwards radial direction. Formally, this spherically symmetric potential can be modeled as

$$U_{\text{sph}}(\mathbf{r}') = U_0 f\left(\frac{r' - R}{R}\right)\Theta(R - r')$$

being included into the Hamiltonian for each gas particle. Here, $U_0 > 0$, $\Theta(x)$ is the Heaviside function and $f(x) \approx -x$ for $x \approx 0$. This way, when $U_0 \to +\infty$, we re-obtain the rigid wall scenario and, in that limit the interior details of the potential will not matter. Within this framework, answer the following questions:



2.1. [5 points] As the walls of the central sphere are not perfectly rigid, the particles are allowed to penetrate it. When doing so, each will feel a force $\mathbf{F}(\mathbf{r}_i')$ propelling it outwards ($\mathbf{r}_i'$ being the position of the particle i in the coordinate system centered in the sphere). Find an expression for this force, in terms of the $U_{\text{sph}}(\mathbf{r}')$ potential.

2.2. [5 points] Using the canonical ensemble, show that the statistical distribution of $\mathbf{F}(r')$ is

$$\mathbf{F}(r') = N \frac{r'^2 \left[-\frac{dU_{\text{sph}}}{dr'}\hat{\mathbf{r}}\right] \exp\left(-\beta U_{\text{sph}}(r')\right) \mathcal{I}(r')}{\int d^{(3)}\mathbf{r}' \exp\left(-\beta\left[U_{\text{sph}}(r') + V(\mathbf{r}_0 + \mathbf{r}')\right]\right)}$$

where $\mathcal{I}(r')$ is a radial function. Determine the form of $\mathcal{I}(r')$ and interpret it.

2.3. [5 points] Assuming the gas to be in thermodynamic equilibrium, what is the physical meaning of the average $\langle \mathbf{F} \cdot \hat{\mathbf{r}}' \rangle$? How does it relate to the pressure felt by the central sphere? Determine the pressure on the surface of the sphere.

2.4. [5 points] The integrals above can be done in the limit of rigid walls ($U_0 \to +\infty$). In this limit, prove that

$$\int_0^\infty dr'\, r'^2 \left[-\frac{dU_{\text{sph}}}{dr'}\right] \exp\left(-\beta U_{\text{sph}}(r')\right) \mathcal{I}(r') = \frac{1}{\beta} R^2 \mathcal{I}(R).$$

2.5. [5 points] Assuming that $V(\mathbf{r})$ is a slowly-varying function at scale of the central sphere. Show that the expression for the pressure simplifies to:

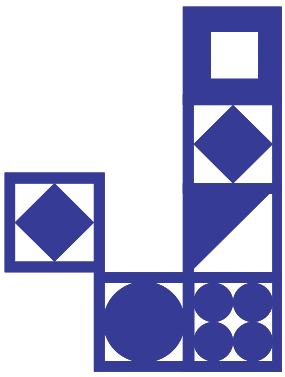
$$P(\mathbf{r}_0) = k_{\text{B}} T \rho(\mathbf{r}_0).$$

Retrieve the ideal gas law. Which confining potential will you use?

2.6. [5 points] Consider the case in which the sphere is centered on an harmonic well potential – $\mathrm{V}(r) = \frac{1}{2}\mathrm{k}|r|^2$. Obtain the exact expression of the pressure exerted on the sphere as a function of temperature and k. Derive the ideal gas law in the limit $k \to 0^+$ and the first correction due to a finite k. Physically justify the sign of this first correction. [**Hint**: Despite there being no container, the confining potential allows one to define an effective volume, where the gas gets concentrated in equilibrium.]

3. A Statistical Derivation of Archimedes' Principle

Since the classical antiquity, it is known that a solid body immersed in a fluid is acted upon by a vertical Buoyancy force which precisely equals the weight of the displaced volume of fluid (*Archimedes' Principle*). Using the framework devised above, one can re-obtain this result using a microscopic theory.



3.1 [10 points] Derive the average force in the $\hat{\mathbf{u}}$ direction $\langle \mathbf{F} \cdot \hat{\mathbf{u}} \rangle = \bar{F}_u$ caused by the gas **on the sphere**, i.e.

$$\bar{F}_u = -N \frac{\int_0^\infty dr'\, r'^2 \left[-\frac{dU_{\text{sph}}}{dr'} \right] \exp\left(-\beta U_{\text{sph}}(r') \right) \mathcal{J}(r')}{\int d^{(3)}\mathbf{r} \exp\left(-\beta V(\mathbf{r}) \right)},$$

where $\mathcal{J}(r' = |\mathbf{r} - \mathbf{r}_0|) = \int_0^{2\pi} d\phi' \int_0^{\pi} d\theta' \cos\theta' \sin\theta' \exp\left(-\beta V(r', \theta', \phi') \right)$ by aligning the z' axis of the spherical coordinates along $\hat{\mathbf{u}}$.

3.2. [10 points] If $V(\mathbf{r})$ is considered constant across the sphere, the Buoyancy force is zero. Based on the hydrostatic interpretation of this force, provide a physical justification for this result.

3.3. [10 points] In the rigid wall limit $U_0 \to +\infty$, obtain the following expression for the Buoyancy force:

$$\mathbf{F} = -\frac{4}{3}\pi R^3 \rho(\mathbf{r}_0)\left(-\boldsymbol{\nabla} V(\mathbf{r}_0) \right).$$

If $V(\mathbf{r})$ is an uniform gravitational potential, show that the previous expression reduces to Archimedes' principle

$$F_{\text{A}} \quad = \quad V_b \rho(z) m g$$

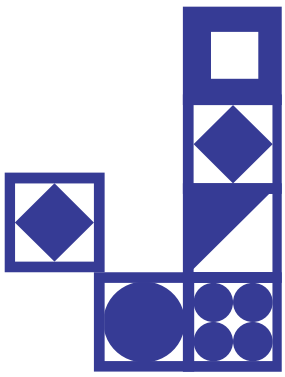
where V_b is the volume of the sphere.

[**Hint:** Approximate $\exp\left(-\beta V(\mathbf{r}) \right)$ by a multivariable Taylor series around $\mathbf{r}_0$ (up to first order) and express it in spherical coordinates. Set $\hat{\mathbf{u}}$ as the direction of the gradient of V]

4. How can a Soap Bubble be used a Barometer?

Previously, we have used a small spherical probe as a theoretical device to define the pressure at any point inside a gas, in a statistical sense. Now, we will see that such device can actually be put to practice and used as a local barometer.

Consider a soap bubble of radius R suspended in the gas. This bubble is stable thanks to the mutual equilibrium of its internal gas pressure, the surface tension of the thin soap film and the pressure due to the external gas. If the external pressure is increased, the radius of the bubble must decrease accordingly, in order to achieve a new equilibrium state. Therefore, the radius R can be used as an indirect measure of the local pressure of the gaseous environment. The change in energy due to a change in area of a surface held together by surface tension is given by $dE = \gamma dA$, where $\gamma > 0$, so the system tends to smaller areas. Assuming the gas inside the bubble to be an ideal gas in thermodynamic equilibrium, answer the following questions.



4.1. [10 points] Obtain Laplace's equation,

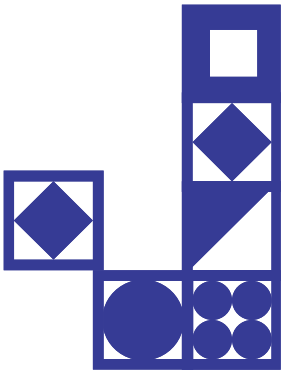
$$P_{out} - P_{in} = \frac{-2\gamma}{R}$$

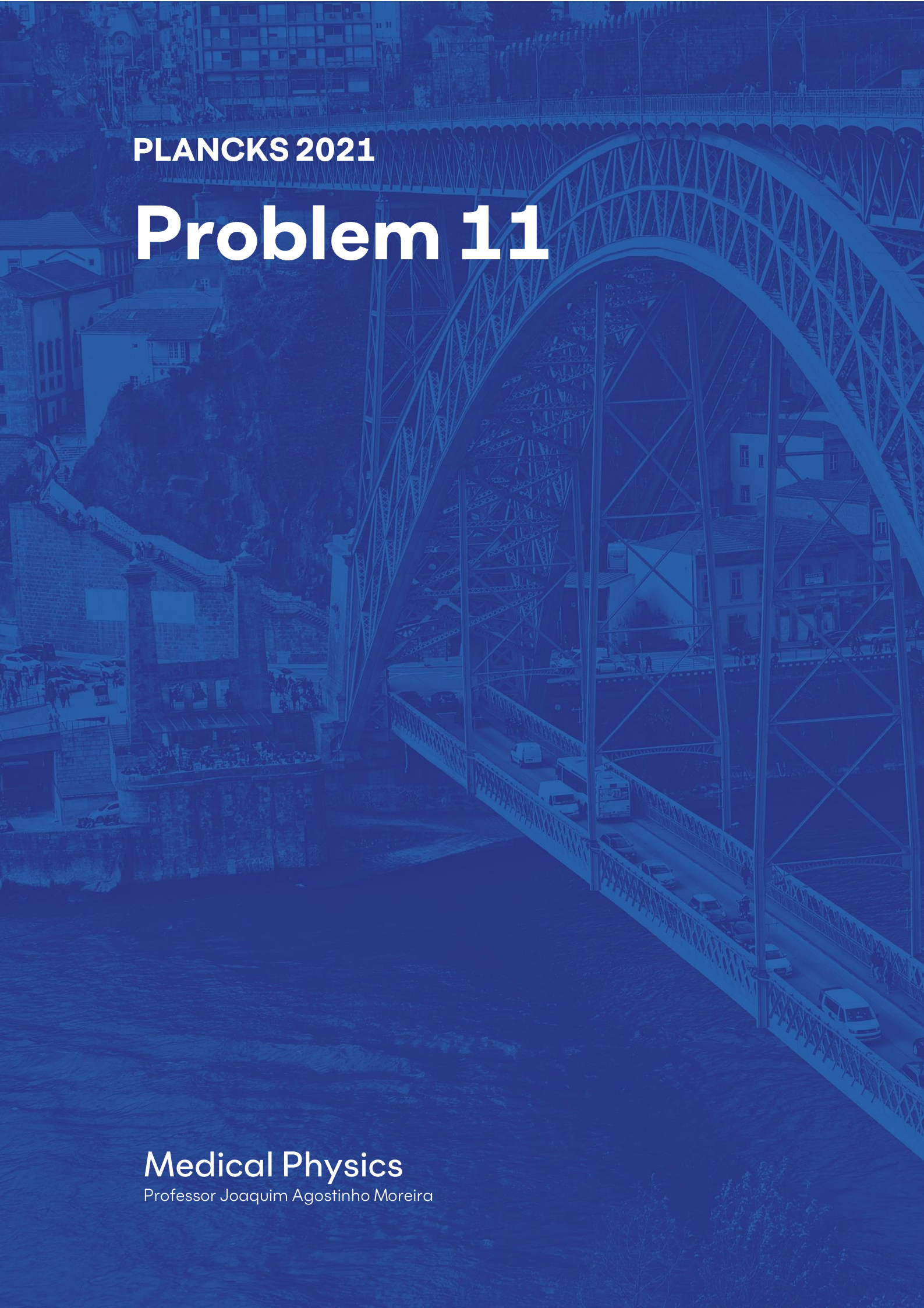
for the equilibrium of forces acting on the soap bubble.

4.2. [10 points] Show that the external pressure is related to the bubble's radius (R) through

$$P_{out} = \frac{Nk_BT}{\frac{4}{3}\pi R^3} + \frac{-2\gamma}{R}$$

where N is the number of gas particles inside the bubble and T is the temperature. Verify that this expression allows for a one-to-one correspondence between R and P_{out}, across any physical regime.





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Problem 11

Medical Physics

Professor Joaquim Agostinho Moreira

Introduction

The Magnetic Resonance phenomenon is the basis of a medical tool used to obtain anatomic and functional images of the human body, widely spread around the world.

To do so, powerful superconducting solenoids or Helmholtz coils produce strong magnetic fields that create a macroscopic magnetization arising from the nuclear magnetic momenta alignment. To obtain suitable signals to build the image, the magnetization must be perturbed from its equilibrium direction, which can be done using radiofrequency pulses. When this perturbation is removed, the magnetization relaxes towards the equilibrium value. During this process, the signal for imaging is recorded.

Consider a proton under the influence of a uniform magnetic field $\vec{B}_o = B_o\hat{z}$. The magnetic moment $\vec{\mu}$ of proton is related with the total angular momentum $\vec{J}$ following the equation $\vec{\mu} = \gamma\vec{J}$, where γ is the gyromagnetic ratio. Assume that no dissipative mechanism is considered and consider a fixed reference frame (x, y, z).

Question 1 [30 points]

Write the equation of motion of the magnetic moment under the influence of the magnetic field, assuming that the angle between these two vectors is θ_0, and show that each magnetic momentum precesses around the direction of the external magnetic field $\vec{B}_o$.

Question 2 [5 points]

Determine the precession frequency ω_o of the magnetic moment.

Question 3 [25 points]

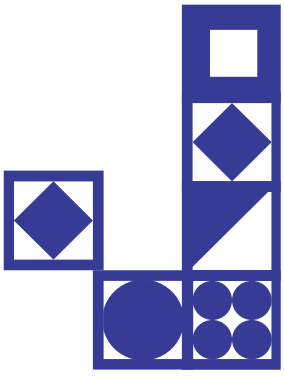
A second reference frame (x', y', z') is now considered which rotates with angular velocity $\vec{\Omega} = -\omega_0\hat{z}$, in such a way that $z = z'$. Write the equation of motion of the magnetic moment referring to the rotating reference frame.

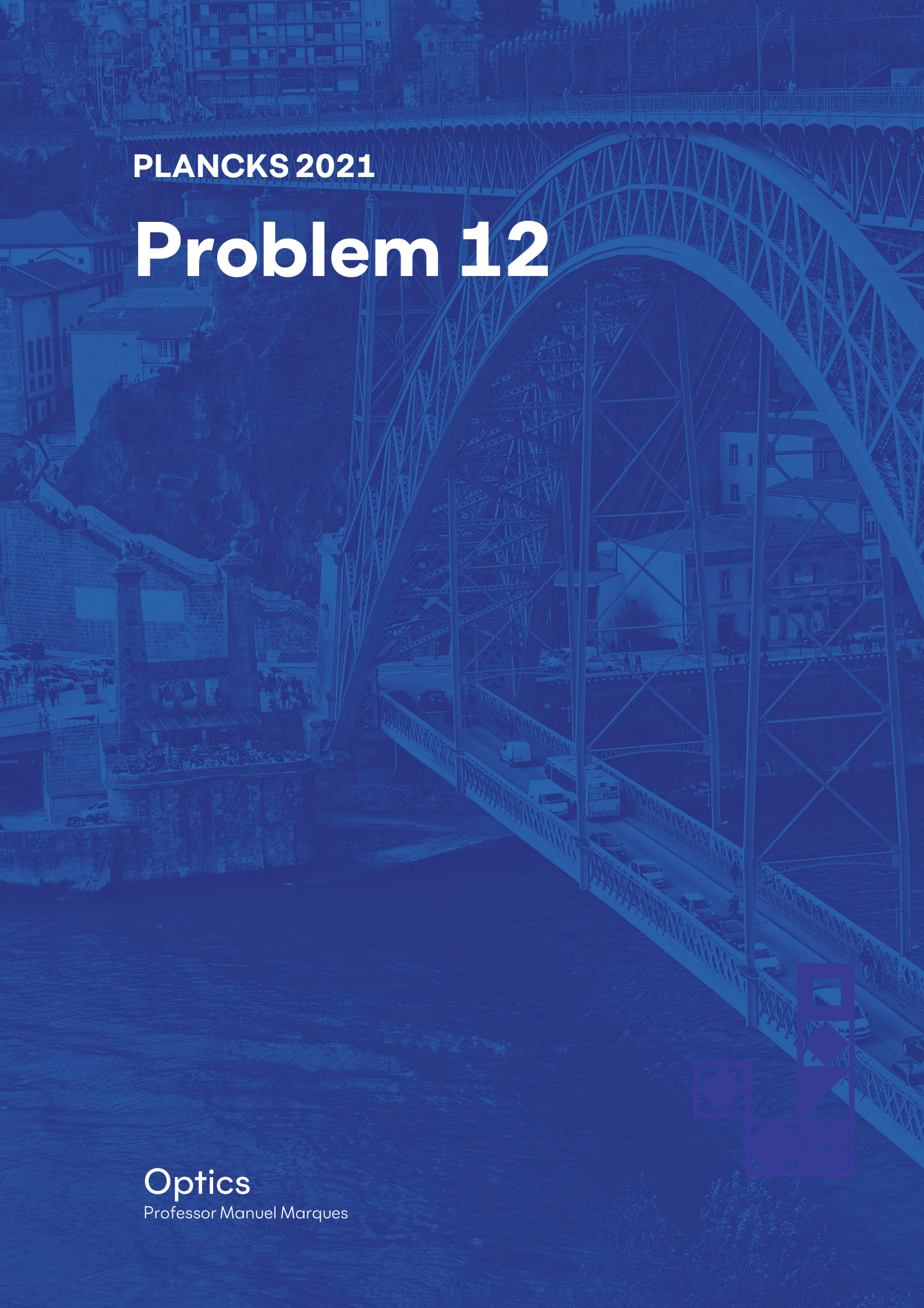
Question 4 [30 points]

Assume now that a known radiofrequency magnetic field described by $\vec{B}_1(t) = B_1(t)\cos(\omega t)\hat{x} - B_1(t)\sin(\omega t)\hat{y}$, with B_1(t) a rectangular function, applied during a time interval τ. Show that the magnetic moment precesses around the x' axis with angular frequency $\omega_1 = \gamma B_1$ if the condition $\omega = \omega_o$ holds. This is the case of resonance conditions.

Question 5 [10 points]

Calculate τ to flip the magnetic moment onto the y'-axis.





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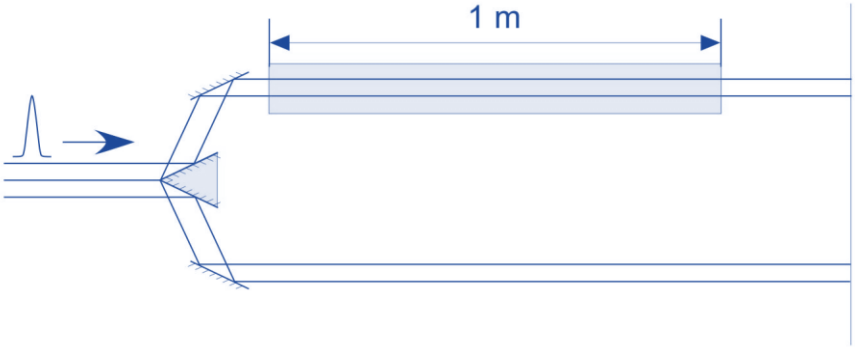
Problem 12

Optics

Professor Manuel Marques

Question 1 [20 points]

A short light pulse ($\lambda_0 = 590nm$) is split in two by a prism mirror. One of the beams travels in the air (refractive index 1.000) and the other travels through one meter of glass (silica) with a refractive index of 1.458.



Compute the time difference of the two pulses arriving at the target.

Note: For simplicity, assume $c = 3.000 \times 10^8$ m/s in all questions of this problem.

Question 2 [20 points]

Let's assume that the two beams in the previous system interfere constructively at the target (mirrors are used to superpose them). What would be the minimum temperature variation of the one-meter silica bar to change the interference from constructive to destructive interference?

The temperature coefficient of the refractive index of silica is 8.7×10^{-6} K^{-1}.

Question 3 [20 points]

A short pulse of light cannot be considered monochromatic. The limitation in the time will truncate the sinusoidal oscillation of the electromagnetic wave. The resulting spectrum depends on the temporal shape of the pulse (the shape that makes the envelope of the ideal electromagnetic sinusoidal wave). For the particular mathematical Gaussian shape envelope, the resulting envelope of the frequencies shape is also a gaussian, and the product of the two Gaussian envelopes is equal to one ($\Delta t \times \Delta f = 1$). This means that a gaussian very short pulse with 10 fs (1 fs = 10^{-15} s) will have a gaussian spectrum with 10^{14} Hz width (corresponding to 83 nm for a pulse centered at 500 nm).

Consider a 300 km fiber optic link with gaussian digital pulses carrying the information. The fiber optic link is operated at the 1550 nm window ($\lambda_0 = 1550$ nm), where the refractive index is 1.468 and has a linear variation with wavelength 5.4×10^{-6} nm^{-1}.

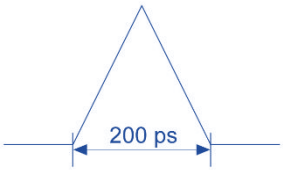
To allow some simplification of the problem, we will assume a gaussian pulse with width τ in a time slot of 2τ (meaning that for a transmission frequency of 1 MHz, τ is equal to half a period of the data frequency, $\tau = 1\mu s/2$), and that the pulse can spread in time by 25% (τ can increase 1.25 times during the propagation).

What would be the maximum frequency of transmission for this simplified system, due to the different speeds of the pulse spectral components?

Note: with these simplifications, the maximum data frequency computed will be smaller than what an optical fiber can carry. (The refractive index variation with wavelength crosses zero at the second transmission window, around 1310 nm).

Question 4 [20 points]

For simplification, consider now a triangular-shaped pulse centered at 1550 nm wavelength, and with a temporal duration of 200 ps.



The pulse carries an energy of 200 nanojoules and has a diameter of 10 micrometers in the optical fiber (for simplification assume a uniform distribution of power).

One of the effects that limit the performance of fiber optical systems is the nonlinear refractive index. The speed of light inside a material can be changed by the irradiance of light (power per unit area), in an optical fiber, this effect has a value of $3.0 \times 10^{-20} \mathrm{m}^2/\mathrm{W}$. In the leading part of the pulse, as the power rises in time, the refractive index increases, decreasing the speed of light (this is similar to the distance to the entrance of the fiber increasing with time). The effect reverses at the back of the pulse. This variation of the velocity shifts the frequencies of the pulse in an analogous way to the Doppler shift.

Compute the new spectral width of this pulse after 1 m propagation, considering this nonlinear effect.

Note: assume for the initial spectrum the one obtained with a Gaussian pulse with the 200 ps width.

Question 5 [20 points]

The dispersion of the fiber at around 1550 nm is anomalous, i.e., the longer wavelengths travel slower than the shorter ones. Since the nonlinear refractive index shifts the wavelengths in the leading edge of the pulse to longer wavelengths, and the ones in the trailing edge to shorter wavelengths, the shifting of wavelengths can be compensated by the different propagation velocities. This is indeed used for the soliton propagation of pulses (a hyperbolic secant shape is one solution for these pulses).

Compute the pulse energy, for a triangular pulse with 200 ps duration (like the one considered in Question 4) that keeps the same pulse duration after 50 km, using the simplifications below.

Notes: As said, this is only possible for a special shape of the pulse, so we will consider a simplified problem. First, we will consider, like in Question 3, that the original pulse spectrum is equal to the one of a Gaussian pulse of 200 ps. Second, we consider that the nonlinear effect is only present at the first meters of the fiber (no nonlinear behavior on the rest of the propagation to separate the two effects). We will consider only the points at half height (separated by 100 ps). We will consider only the maximum shifted frequencies (the slower at the leading edge, and the faster at the trailing edge) for calculation. The equal pulse duration to be computed shall be simplified to be that the slower wavelength generated at the leading edge is 100 ps slower than the faster wavelength generated at the trailing edge.

Note: At 1550 nm the refractive index is 1.468 and has a linear variation with wavelength of 5.4×10^{-6} nm^{-1}. The nonlinear refractive index is $3.0 \times 10^{-20} m^2/W$. Assume a uniform spot with 10 micrometer diameter.

